Master Thesis

Influence of the Exhaustion of the Gluteus Medius on the Knee Adduction Moment During Forward Lunge Exercise

carried out for the purpose of obtaining the degree of

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by

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English Abstract

Introduction: Osteoarthritis (OA) of the knee is a highly prevalent musculoskeletal disorder and a leading cause of pain and functional impairment worldwide. Biomechanical loading of the knee joint, particularly the knee adduction moment (KAM), has been identified as a key factor associated with OA development and progression. Despite its clinical relevance, KAM behaviour during commonly used rehabilitation exercises such as the Forward Lunge (FL) remains insufficiently characterised, particularly in the context of hip abductor fatigue, which has been shown to influence knee joint loading.

Materials & Methods: This study investigated KAM patterns during FL in healthy young male adults and examined whether targeted fatigue of the gluteus medius (GMED) alters knee joint loading. A single-group pre–post experimental design was conducted in a motion analysis laboratory at TU-Wien. Three-dimensional motion capture, ground reaction force measurements, and musculoskeletal modelling using an inverse dynamics approach were applied to derive KAM during FL execution.

Results: It was shown that FL elicited a consistent KAM pattern across participants, with a characteristic peak occurring in the mid portion of the ground-contact phase. Peak KAM values were comparable between pre- and post-fatigue conditions, indicating minimal change in frontal-plane knee loading following the intervention. Fatigue assessment based on surface EMG spectral characteristics did not confirm successful GMED fatigue, as median frequency (MNF) increased rather than decreased following the intervention

Discussion & Conclusion: This thesis provides normative data on KAM, GMED activation and contraction force, as well as kinematic and kinetic characteristics during the FL in healthy young male adults. Contrary to expectations, the applied protocol resulted in reduced KAM alongside increased GMED activation and force, suggesting enhanced muscular engagement rather than fatigue. These findings may support future rehabilitation assessment and progression strategies.

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Deutsche Kurzfassung

Hintergrund: Die Kniearthrose zählt zu den häufigsten muskuloskelettalen Erkrankungen und stellt weltweit eine der Hauptursachen für Schmerzen sowie funktionelle Einschränkungen dar. Die biomechanische Belastung des Kniegelenks, insbesondere das Knieadduktionsmoment (KAM), gilt als ein wesentlicher Faktor für die Entstehung und den Fortschritt der Kniearthrose. Trotz seiner klinischen Relevanz ist das Verhalten des KAM während häufig verwendeter Rehabilitationsübungen wie dem Ausfallschritt nach vorne (FL) bislang nur unzureichend untersucht. Dies gilt insbesondere im Zusammenhang mit einer Ermüdung der Hüftabduktoren, welche nachweislich die Belastung des Kniegelenks beeinflussen kann.

Material & Methoden: Ziel der vorliegenden Arbeit war die Untersuchung des KAM während des FL bei jungen gesunden männlichen Erwachsenen sowie die Analyse eines möglichen Einflusses einer gezielten Ermüdung des Musculus gluteus medius (GMED) auf die Kniegelenksbelastung. Hierfür wurde ein experimentelles Prä-Post-Design mit einer Untersuchungsgruppe im Bewegungsanalyselabor der TU-Wien durchgeführt. Zur Bestimmung des KAM wurden dreidimensionale Bewegungsanalysen, Bodenreaktionskraftmessungen sowie muskuloskelettale Modellierungen mittels inverser Dynamik eingesetzt.

Ergebnisse: Der FL zeigte über alle Probanden hinweg ein konsistentes KAM-Profil mit einem charakteristischen Maximum im mittleren Abschnitt der Bodenkontaktphase. Die maximalen KAM-Werte waren vor und nach der Ermüdungsintervention vergleichbar, was auf nur geringe Veränderungen der Kniegelenksbelastung in der Frontalebene hinweist. Die Analyse der spektralen Eigenschaften der EMG-Signale bestätigte keine erfolgreiche Ermüdung des GMED, da die Medianfrequenz nach der Intervention anstieg, anstatt abzunehmen.

Schlussfolgerungen: Diese Arbeit liefert normative Daten zum KAM, zur Aktivierung und Kontraktionskraft des GMED sowie zu kinematischen und kinetischen Charakteristika während des FL bei gesunden jungen Männern. Entgegen der Erwartung führte das angewandte Protokoll zu einem reduzierten KAM bei gleichzeitig erhöhter Aktivierung und Kraft des GMED, was eher auf eine verstärkte muskuläre Beanspruchung als auf Ermüdung hindeutet. Die Ergebnisse können zur Weiterentwicklung evidenzbasierter Rehabilitations- und Progressionskriterien beitragen.

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List of Abbreviations

ACL ..... Anterior Cruciate Ligament

ACLR ..... Anterior Cruciate Ligament Reconstruction

ASIS ..... Anterior Superior Iliac Spine

AST ..... Automatic Scaling Tool

BMI ..... Body Mass Index

DOF ..... Degree of Freedom

EHIS-PAQ ..... European Health Interview Survey - Physical Activity Questionnaire

EMG ..... Electromyography

FE ..... Fatigue Exercise

FL ..... Forward Lunge Exercise

GMED ..... Gluteus Medius Muscle

GRF ..... Ground Reaction Force

HTO ..... High Tibial Osteotomy

ID ..... Inverse Dynamics

IK ..... Inverse Kinematics

JRA ..... Joint Reaction Analysis

KAM ..... Knee Adduction Moment

KJCF ..... Knee Joint Contact Force

MA ..... Moment Arm

MNF ..... Median Frequency

MVIC ..... Maximum Voluntary Isometric Contraction

NSAID ..... Non-Steroidal Anti-Inflammatory Drugs

OA ..... Osteoarthritis

PSD ..... Power Spectral Density

PSIS ..... Posterior Superior Iliac Spine

PTOA ..... Post Traumatic Osteoarthritis

RMS ..... Root Mean Square

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ROM ..... Range of Motion

RRA ..... Reduced Residual Algorithm

sEMG.....Surface Electromyography

SENIAM .....Surface Electromyography for the Non-Invasive Assessment of Muscles

SO.....Static Optimisation

vGRF .....vertical Ground Reaction Force

XML ..... Extensible Markup Language

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1 Introduction

1.1 Motivation

Research on biomechanics of the human knee is of substantial clinical relevance, as studies have shown, that biomechanical metrics are directly linked to symptoms perseveration and disease progression in patients with musculoskeletal disorders. In particular, research on the type and intensity of joint loading revealed a connection to clinical symptoms in patients. Sufferers, affected from knee OA for instance, were shown to exhibit elevated joint compression, influenced by muscle contraction forces, highlighting the importance of biomechanical assessments in order to understand patient outcomes (Aljawadi, 2019).

Among commonly studied biomechanical parameters, medial knee loading — which has been shown to be closely connected to medial knee OA — has received considerable attention. The knee adduction moment (KAM), frequently used as a surrogate measure for medial compartment loading, has been consistently associated with clinical symptoms. Specifically, higher medial knee loading is linked to increased knee pain during walking, which highlights its high relevance for function and diagnostics (Birmingham et al., 2019).

Beyond knee-specific parameters, proximal joint mechanics were also shown to contribute to disease progression. For instance, greater internal hip abduction moments have been shown to exert a protective effect against the progression of medial tibiofemoral OA (Chang et al., 2005). These findings are in alignment with others, which discovered a correlation between knee pain and a significantly weaker isometric hip abductor strength profile (Deasy et al., 2016) as well as decreased levels of activation of the latter muscle group during functional tasks (Mirzaie et al., 2019).

Recent evidence supports the importance of early intervention based on results of biomechanical assessments. Addressing knee dysfunction at an early stage may help to impede or slow disease progression, thereby underscoring the importance of early detection, characterisation, and quantification of functional impairments to effectively inform therapeutic strategies (Mündermann et al., 2024; Potti et al., 2024). In this context, research has increasingly focused on identifying biomechanical indicators and modifiable risk factors, alongside developing interventions aimed at slowing or halting disease progression (Hutchison et al., 2023).

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1.2 Biomechanical Analysis

Biomechanical analysis refers to the quantitative, often computer-assisted evaluation of forces, moments, and resulting motions within biological systems, particularly the musculoskeletal system. It integrates experimental measurements (e.g., motion capture, force plates, electromyography) with computational modelling approaches such as inverse dynamics (ID) and musculoskeletal simulations to estimate internal joint loads and muscle forces that are not directly measurable in vivo. This approach enables a detailed assessment of movement mechanics and provides insight into the mechanical factors underlying normal function, injury mechanisms, and disease progression (Robertson et al., 2013).

To ensure reliable simulation outcomes, the formulation of an appropriate biomechanical model is essential. Such a model represents a comprehensive description of both the form and function of a biomechanical system, encompassing neural, muscular, and skeletal structures, and, where relevant, incorporating non-biological components such as exoskeletons. Biomechanical models are defined by their constituent elements — typically rigid bodies and musculotendon units — their associated properties (e.g., segment masses and inertial characteristics), and the interactions and constraints governing these components. Their architectural and mechanical properties might be alterable, thereby enabling the reliable representation of specific anatomical configurations and subject-specific characteristics (Seth et al., 2018).

1.2.1 Knee Adduction Moment (KAM)

The KAM is a biomechanical metric used to characterise load distribution between the medial and lateral compartments of the knee joint (Mündermann et al., 2024) and is the most widely used dynamic parameter to represent medial knee joint loading during functional activities such as walking (Liu et al., 2024; Tang et al., 2022).

Different approaches have been proposed for estimating the KAM, resulting in several definitions within the literature. A simple and intuitive approach, commonly referred to as the *lever-arm method*, defines the KAM as the frontal-plane component of the cross product between the spatial vector extending from the knee joint centre to the line of action of the GRF *with* the GRF vector (**Error! Reference source not found.**). Conceptually, this approach quantifies the moment generated by the GRF about the knee joint centre and can be calculated with comparatively limited data and processing requirements.

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Consequently, it may represent a practical option when a rapid estimate of frontal-plane knee loading is desired (Lewinson et al., 2015).

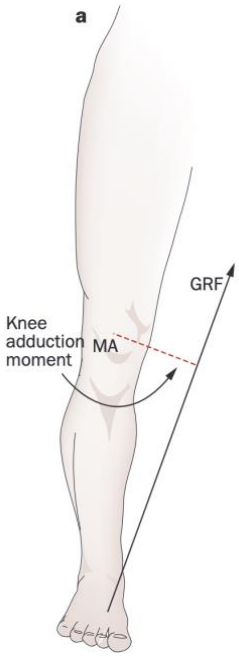


Fig. 1: One common conceptual definition of the KAM:
GRF acting through a moment arm (MA) relative to the knee joint centre
reprinted from Reeves & Bowling (2011)

A more comprehensive approach derives the KAM from the equations of motion using ID. Based on Newton–Euler mechanics, this method determines the net joint moments required to generate the observed movement while accounting for three-dimensional segment kinematics, external forces, body-segment inertial properties, and gravitational effects. As such, the resulting KAM reflects not only the geometric relationship between the knee joint centre and the GRF, but also the dynamic behaviour of the lower-limb segments (Lewinson et al., 2015).

The present thesis adopts this more comprehensive approach (Lewinson et al., 2015) based on equations-of-motion for the definition of the KAM: Following ID, muscle forces are estimated and subsequently incorporated into the calculation of joint reaction loads (Hicks, 2019a). Consequently, the KAM reported in the current thesis corresponds to the frontal-plane component of the knee joint moment transferred between the femur and tibia as a result of all loads acting on the model, including contributions from muscle forces and external loading conditions. Within this framework, the KAM represents the portion of

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frontal joint moment not balanced by muscle forces and is transferred solely by passive joint structures, including articular surfaces and ligaments.

As outlined in Section 1.3.1 Surgical Treatment, the KAM is strongly influenced by the mechanical axis of the lower limb, which may be modified through surgical interventions. A common pattern observed is that knees with varus malalignment typically exhibit elevated KAM values (Van Egmond et al., 2017). However, KAM is not solely determined by structural factors; functional aspects also play a significant role. In particular, hip abductor musculature contributes to frontal-plane knee joint mechanics, with reduced hip abductor strength being associated with elevated KAM values (Geiser et al., 2010).

Substantial evidence indicates that elevated KAM is associated with both the onset and progression of medial knee OA, as reflected by structural changes such as cartilage defects (D'Souza et al., 2022) and thinning (Chehab et al., 2014), bone marrow lesions, joint space narrowing (D'Souza et al., 2022) and meniscal tears (Davies-Tuck et al., 2008). In addition to structural degeneration, the KAM is as well associated with clinical symptoms, particularly pain. Multiple studies, including meta-analytical evidence, report a positive relationship between KAM and pain experienced during gait (Hutchison et al., 2023; Nie et al., 2019). Conversely, lower frontal plane loading metrics, hence, reduced KAM values, have been associated with more favourable long-term outcomes in post-traumatic knee populations, suggesting a prognostic value of this parameter in case of presence of a pathology (Savage et al., 2025).

Despite its strong associations with clinical symptoms and structural degeneration, the explanatory power of the KAM has also been questioned. In particular, the KAM has been criticised for being interpreted as a direct surrogate of medial knee loading, although it more accurately reflects the distribution of load between the medial and lateral compartments rather than absolute joint loading itself (Mündermann et al., 2024).

1.2.2 Gluteus Medius Muscle (GMED)

Hip abductor musculature — with particular emphasis on the GMED (Fig. 2) as the primary muscular stabiliser — plays a central role in securing the lower limb in the frontal plane and contributes to the control of femoral motion (Mirzaie et al., 2019; Tang et al., 2022). The GMED is essential for maintaining pelvic stability during single-leg stance, ensuring step-to-step symmetry during gait, and supporting trunk stabilisation and limb placement during functional tasks (Tang et al., 2022; Thomas et al., 2022). Consequently, the muscle

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is a key modulator of frontal-plane knee joint mechanics and indirectly influences knee joint loading including metrics such as the KAM (Geiser et al., 2010).

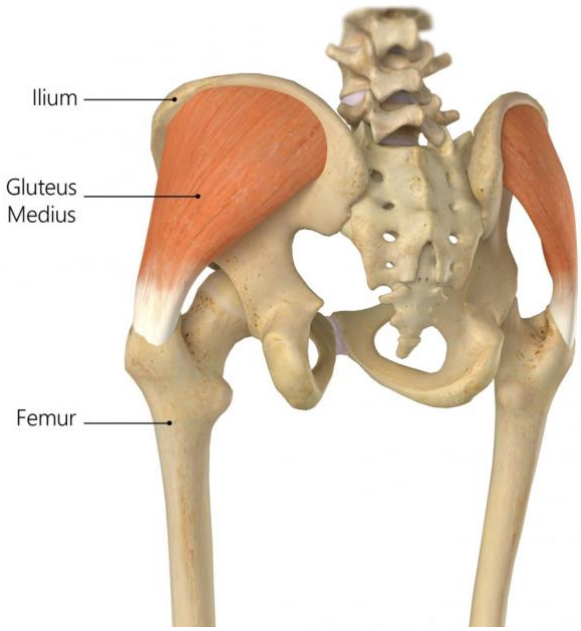


Fig. 2: Anatomical location of the GMED relative to the pelvis, femora, and lower lumbar spine reprinted (Gluteus Medius, 2020)

GMED function plays a critical role in linking biomechanical variables to clinical outcomes in knee joint pathologies. During gait, insufficient hip abductor strength may lead to contralateral pelvic drop during the stance phase, thereby shifting the body's centre of mass laterally and increasing its moment arm relative to the knee joint, which consequently elevates the KAM (Thomas et al., 2022). Experimental evidence further supports a causal relationship between hip abductor fatigue and increased KAM during functional tasks (Geiser et al., 2010). Accordingly, recent studies have attempted to experimentally induce GMED fatigue as a model of transient hip abductor insufficiency (Dimitriou et al., 2025). In this context, electromyographic MNF, defined as the MNF of the EMG power spectrum, has been used as a reliable parameter for quantifying muscular fatigue (Mohseni Bandpei et al., 2014). Conversely, adequate hip abductor torque during the stance phase of gait contributes to pelvic stabilisation and maintenance of physiological load distribution across the knee joint (Thomas et al., 2022). This is clinically relevant, as patients with knee OA generally exhibit elevated KAM values (Valente et al., 2025) and reduced physical function due to symptoms like pain compared to healthy individuals (Nie et al., 2019).

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Importantly, targeted strengthening of the GMED has been shown to reduce pain and improve functional outcomes in patients with knee OA, underscoring its clinical relevance as an important factor in rehabilitation (Thomas et al., 2022).

1.2.3 Forward Lunge Exercise (FL)

The FL (Fig. 3) is a highly adaptable functional strengthening exercise which is being extensively used in rehab and prevention (Begalle et al., 2012) as well as in the fields of sports as a preparational or performance enhancing strengthening exercise (Bishop et al., 2017). It is this thesis' central exercise to be examined.

There are many different variations in the execution of the forward lunge exercise which itself is already a variation. It can be executed varying the trunk position (Bezerra, 2021; Farrokhi et al., 2008), the step length and stride variation (Escamilla et al., 2025) as well as other exercise determinants as e.g. using a step or not (Escamilla et al., 2022), or using additional weight or not (Gao et al., 2022).



Fig. 3: Person during the performance of a FL in the lowest position of the movement cycle reprinted from Begalle et al., (2012)

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1.2.4 OpenSim

One software, enabling the simulation of biomechanics is OpenSim (Delp et al., 2007). OpenSim is an open-source software platform designed to model, simulate, and analyse the neuromusculoskeletal system. It is developed and continuously maintained by a growing research community via *SimTK*[1], a public repository that provides musculoskeletal models, experimental data, and simulation tools to support physics-based investigations of biological systems (Delp et al., 2007). As a community-driven platform, both the development of new models and the interpretation of clinically relevant results are substantially shaped by contributions from OpenSim users (Seth et al., 2018).

Functionally, OpenSim operates as an integrated computational engine that streamlines biomechanical analyses through a series of modular components (Butler et al., 2017), including models of muscles, joints and other musculoskeletal structures (Seth et al., 2018) (Fig. 4). Its intuitive GUI can be extended and customised using extensible markup language (XML), allowing users to adapt and expand the software's functionality to specific research applications (Butler et al., 2017).

The software's computational models of the human body and its substructures are grounded in scientific data accumulated over decades of research. To ensure their reliability and scientific validity, these models — as well as their simulation outputs — are continuously evaluated through validation experiments and comparisons with independent models (Seth et al., 2018).

OpenSim provides a versatile environment for the development, modification, and analysis of musculoskeletal models, in which muscles, joints, and other anatomical structures can be parameterised in terms of their architecture and mechanical properties to represent specific systems of interest. Central to its functionality is the ability to construct, manipulate, and interrogate biomechanical models, enabling detailed investigation of movement mechanics. Building upon this, OpenSim facilitates the simulation of musculoskeletal dynamics and neuromuscular control, which is particularly valuable in contexts where direct experimental measurement is challenging or infeasible, such as

[1] https://simtk.org/

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quantifying internal joint contact forces during dynamic tasks like walking. Finally, OpenSim offers a graphical user interface that supports the interpretation, verification, and communication of simulation results (Seth et al., 2018).

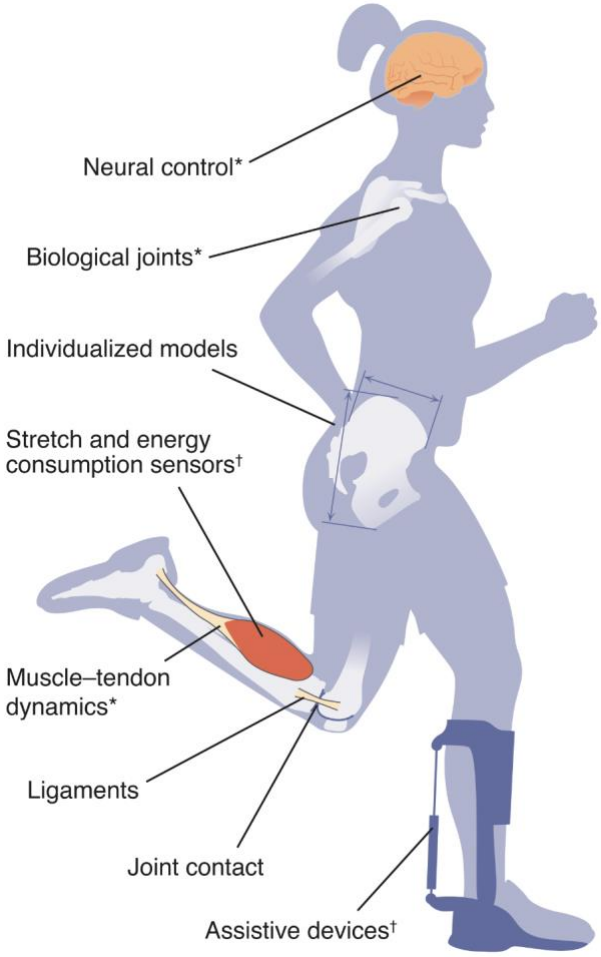


Fig. 4: Overview of the modelling capabilities of OpenSim, including neural control, musculoskeletal dynamics, joint mechanics, and assistive devices reprinted from Seth et al., (2018)

A key strength of OpenSim lies in its ability to integrate experimental motion capture data into a computational modelling framework, thereby enabling subsequent biomechanical analysis. This is achieved through the representation of reflective markers within a three-dimensional virtual environment, which establishes the link between experimentally recorded kinematics and the musculoskeletal model. To ensure an accurate correspondence between virtual and experimental marker trajectories, OpenSim employs a least-squares optimisation approach that minimises the discrepancy between both marker sets (Butler et al., 2017).

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Within the present thesis, OpenSim was used as the central computational framework for musculoskeletal modelling and model-based biomechanical analysis. Specifically, its core capabilities in inverse kinematics (IK), inverse dynamics (ID), residual reduction algorithm (RRA), static optimisation (SO), and joint reaction analysis (JRA) were utilised to transform experimental motion capture and Ground Reaction Force (GRF) data into physiologically meaningful biomechanical quantities. The software enabled the generation and processing of subject-specific models, thereby allowing the estimation of internal joint loading and muscle forces. OpenSim was thereby used within a semi-automated MATLAB-based (The MathWorks Inc, 2024) workflow. Consequently, OpenSim served as the key tool linking experimental data with computational modelling to be able to quantify the KAM and thereby fulfilling this thesis' goals.

In the following subchapters, the single Tools of an inverse biomechanical motion analysis approach will be explained in detail.

1.2.4.1 Scaling

Scaling is the process of adapting the properties of a generic human body model optimally to the specifics of the subject of interest (Hicks, 2019b).

The process of scaling a generic musculoskeletal human model is a crucial one in general (Hicks, 2019b), and therefore also for the outcome of this thesis, because even small mismatches may substantially affect intermediate and final results of the post-processing pipeline (Butler et al., 2017).

Generally, the scaling of a model is usually performed by means of comparing distances between meaningful markers — experimental or virtual — and corresponding distances of the generic model. Putting these pairs of distances in ratios results in scale-factors which in the following step are applied to the generic model during the actual process. While scaling of rather complex bony bodies such as the pelvis or the torso may be done in all three coordinate directions separately, scaling of long bone segments like the thigh is agreed to be sufficient using two scale factors only — one for the long axis and one for the transversal direction. This is known as the best practice method according to the chapter „Scaling“ (2019) of the official OpenSim Documentation (Hicks, 2019b).

The RajagopalLaiUhlrich2023 model

The *RajagopalLaiUhlrich2023* musculoskeletal model (Lai et al., 2017; Rajagopal et al., 2016; Uhlrich et al., 2022) is a common biomechanical full-body model which offers

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6-Degree-of-Freedome modelling of the knee joint. The model represents an extension of the original full-body model developed by Rajagopal et al. (2016). The original model comprises detailed full-body bony geometry, 37 degrees of freedom, 80 Hill-type muscle–tendon units actuating the lower extremities, and ideal torque actuators controlling the upper body. It was primarily developed and validated for muscle-driven simulations of healthy gait. However, research identified limitations of the model when simulating movements involving substantial hip and knee flexion, as excessive passive muscle forces could lead to unrealistic antagonist co-activation patterns. Lai et al. (2017) and Uhlrich et al. (2022) refined the original model of Rajagopal et al. (2016) by increasing the permissible range of knee flexion, updating muscle paths around the knee joint, and modifying the force-generating properties of selected muscles. The refined model demonstrated improved reliability of muscle activations and reduced excessive passive joint moments and antagonist co-activation. The updated model is therefore considered suitable for analysing movements involving larger ranges of motion, permitting simulations of up to approximately 120° of hip flexion and 140° of knee flexion.

1.2.4.2 Inverse Kinematics (IK)

The IK tool implemented in OpenSim determines the model pose, that best matches experimentally recorded marker trajectories for each frame of motion data. To achieve this, the tool iteratively positions the musculoskeletal model such that the discrepancy between experimental markers and corresponding virtual model markers is minimised. More specifically, IK solves an optimisation problem by minimising the weighted sum of squared errors between marker positions and, if applicable, coordinate values. As a result, the coordinates of all degrees of freedom (DOF) are determined frame-by-frame over the entire movement sequence. Accurate IK solutions are essential, as they constitute the basis for subsequent biomechanical analyses and modelling procedures within OpenSim, including inverse dynamics and muscle force estimations (Hicks, 2018b).

1.2.4.3 Inverse Dynamics (ID)

The ID tool implemented in OpenSim allows for the calculation of net joint moments and net joint forces required to generate a given movement pattern. Based on classical Newtonian mechanics ($F = ma$), the tool solves the equations of motion in an inverse dynamics approach to determine the generalised forces acting at each joint throughout the movement. For this purpose, the ID tool utilises the kinematic solution obtained from

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IK, external force data such as GRF, and a subject-specific scaled musculoskeletal model. Calculations are performed frame-by-frame over the entire movement sequence. In addition to joint moments and forces, the ID analysis also computes residual forces and moments acting at the pelvis-ground joint, which represent compensatory quantities required to dynamically reconcile model motion and external loading conditions (Hicks, 2018a).

1.2.4.4 Reduced Residual Algorithm (RRA)

The RRA is an optimisation-based tool implemented in OpenSim that aims to improve the dynamic consistency between experimentally derived kinematics and measured GRFs. To achieve this, RRA iteratively adjusts the model's mass distribution and kinematic trajectories in order to minimise dynamic inconsistencies and thereby enhance the physical plausibility of the simulation (Hicks, 2018c).

In the presence of modelling inaccuracies and experimental errors — such as soft tissue artefacts, scaling inaccuracies, or measurement noise — non-physical compensatory forces and moments, referred to as residuals, arise. These residuals reflect inconsistencies between inertial effects, external forces, gravity, and the model's predicted motion, and are applied at the first free joint — the ground-pelvis-joint (Hicks, 2018c).

During the RRA process, the original muscle actuators of the input model are typically replaced by idealised joint torque actuators controlling each degree of freedom. This simplification allows the algorithm to focus on resolving inconsistencies in the system dynamics without the additional complexity of muscle force estimation.

1.2.4.5 Static Optimisation (SO)

SO is a built-in tool in OpenSim, which calculates for muscle force estimates and thereby extends ID by resolving net joint moments into individual muscle forces at each time step. This is achieved by solving an optimisation problem in which the sum of squared muscle activations is minimised (Hicks, 2018d). Muscle force is thereby computed as a function whose argument among others is the product of muscle activation —typically bounded between 0 and 1—and the respective maximum isometric force of each muscle.

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1.2.4.6 Joint Reaction Analysis (JRA)

JRA is implemented in OpenSim as part of the Analysis toolset and requires a complete set of simulation inputs, including a musculoskeletal model, kinematic data, external loads, and muscle force profiles. Based on these inputs, JRA computes the resultant forces and moments acting within joints. More specifically, JRA determines the forces and moments transmitted between adjacent body segments as a consequence of all loads applied to the model, including inertial effects, external forces, and actuator contributions. These quantities represent the internal joint loading required to maintain the prescribed kinematics and are interpreted as the net effect of all modelled and unmodelled joint structures, such as cartilage contact and omitted passive elements, including ligaments. The joint reaction forces are typically expressed at the joint centre of the corresponding articulating bodies (Hicks, 2019a).

In contrast to ID, which computes net generalised forces and moments at each degree of freedom based solely on kinematics and externally applied forces, JRA accounts for the contributions of muscles, motors, and all other actuators included in the model. Consequently, while ID provides net joint moments required to reproduce motion, JRA yields physiologically more representative estimates of internal joint loading by incorporating the full set of force-generating elements (Hicks, 2019a).

The estimation of joint reaction forces requires a complete set of inputs, including the musculoskeletal model, kinematic data, external loads, and muscle force profiles.

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1.3 Osteoarthritis (OA)

Pathophysiology

OA is a multifactorial disease, affecting an entire joint, and is arising from the interplay of trauma, altered mechanical loading, inflammatory processes, biochemical reactions, and metabolic dysregulation. A central feature of OA pathophysiology (Fig. 5) is a state of chronic, low-grade inflammation of joint tissue, predominantly mediated by innate immune mechanisms. These mediators stimulate the production of hydrolytic enzymes, which degrade essential cartilage components such as proteoglycans and collagen. In most of the cases this is ultimately leading to progressive cartilage breakdown — a hallmark of OA — and consecutively in whole-joint degeneration and functional impairment accompanied by pain (Mora et al., 2018).

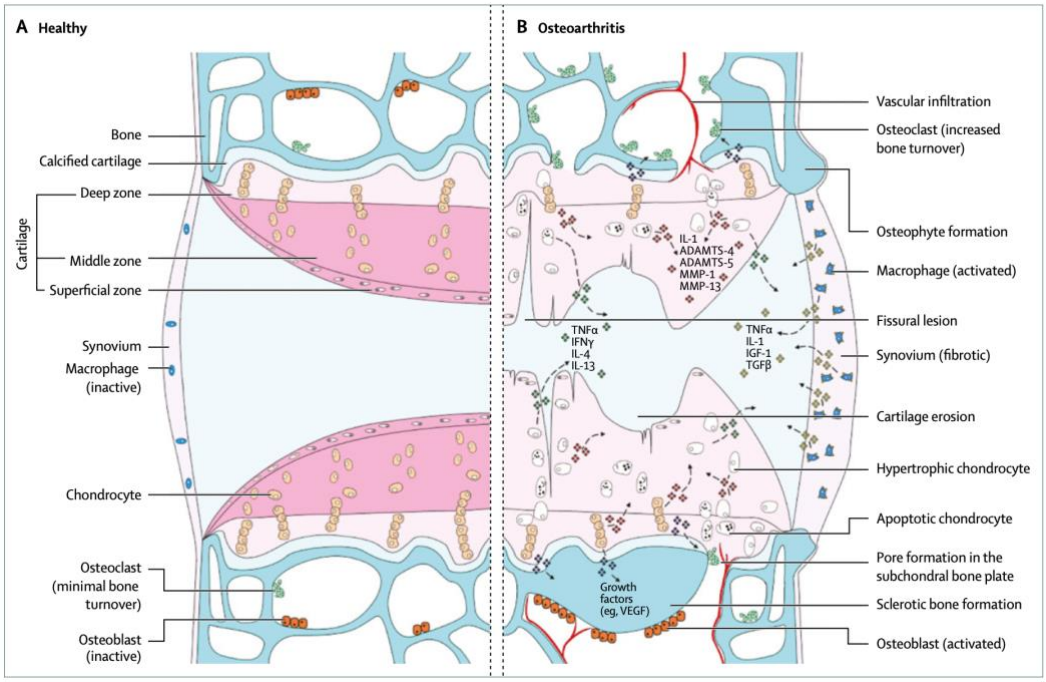


Fig. 5: Comparison of healthy and osteoarthritic joint tissue illustrating key signalling pathways and structural alterations in OA
Reprinted from Hunter & Bierma-Zeinstra (2019)

Demographics and Global Burden

OA is the most common joint disease in adults around the world (Aljawadi, 2019) with 607 million people worldwide suffering from it in the year of 2021 (Qiao et al., 2025). The disease represents a major global public health concern, as its high prevalence and disabling nature already impose a substantial burden to healthcare systems worldwide

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with projections for the next decades even indicating a continuous increase in occurrence (Qiao et al., 2025).

Among other joints, the knee joint is the most common affected one by OA (Englund, 2010), resulting in chronic pain, reduced quality of life, limited physical function, mental health deterioration and significant economic costs (Hutchison et al., 2023).

Influencing Parameters

Despite this global need to address this challenge, the relationship between biomechanical factors — such as innate biomechanical variations, modulators like body weight, injury, pain, or footwear — and their effects on biological processes — tissue's metabolism — in OA risk, development, aetiology, and treatment still remains poorly understood (Mündermann et al., 2024).

Nonetheless, several key associations have been well established in the literature. The onset and progression of knee OA are closely linked to the mechanics of ambulation (Aljawadi, 2019). In particular, KAM during gait have been consistently associated with the development and progression of medial tibiofemoral OA (D'Souza et al., 2022; O'Connell et al., 2016), with body mass index (BMI) acting as a moderating factor in this relationship (Hutchison et al., 2023).

Sustaining a significant knee trauma, such as an ACL injury, represents another important contributing factor, as altered knee biomechanics have been observed both in ACL-deficient and reconstructed knees (Murrell-Smith et al., 2025). These individuals typically exhibit altered frontal-plane kinematics and compensatory movement strategies, which may increase the risk of developing post-traumatic osteoarthritis (PTOA) of the knee joint (Murrell-Smith et al., 2025). Following ACL reconstruction (ACLR), further alterations in knee joint mechanics have been documented, including changes in contact mechanics, increased external tibial rotation, and greater anterior tibial translation during gait (Pandy et al., 2025). Collectively, these biomechanical alterations may contribute to abnormal joint loading and represent potential mechanisms underlying the development and progression of PTOA following ACL injury and reconstruction (Pandy et al., 2025).

In addition to joint loading characteristics, impairments in muscular function have as well been reported, as individuals with symptomatic knee OA demonstrate significantly reduced isometric and isokinetic hip abductor strength compared to healthy controls (Deasy et al., 2016).

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Addressing existing gaps in the understanding of OA pathophysiology is essential to enable effective prevention, diagnosis, and treatment strategies. In this context, biomechanics plays a central role in clarifying the formerly stated unknows about OA which has been consistently emphasised in the scientific literature over many years (Mündermann et al., 2024).

Current Treatment Approachs in Patient Care

Knee OA is a common condition which is currently being either treated by means of a variety of surgical techniques or conservatively (Aljawadi, 2019).

1.3.1 Surgical Treatment

One possible surgical intervention to treat frontal plane knee dysfunction might be surgical high tibial osteotomy (HTO). Thereby, the surgeon performs a change in alignment by inducing a varus or valgus correction of the knee. This surgical technique might be indicated in case of frontal knee malalignment in patients suffering from unicompartmental knee OA (Van Egmond et al., 2017) (Fig. 6).

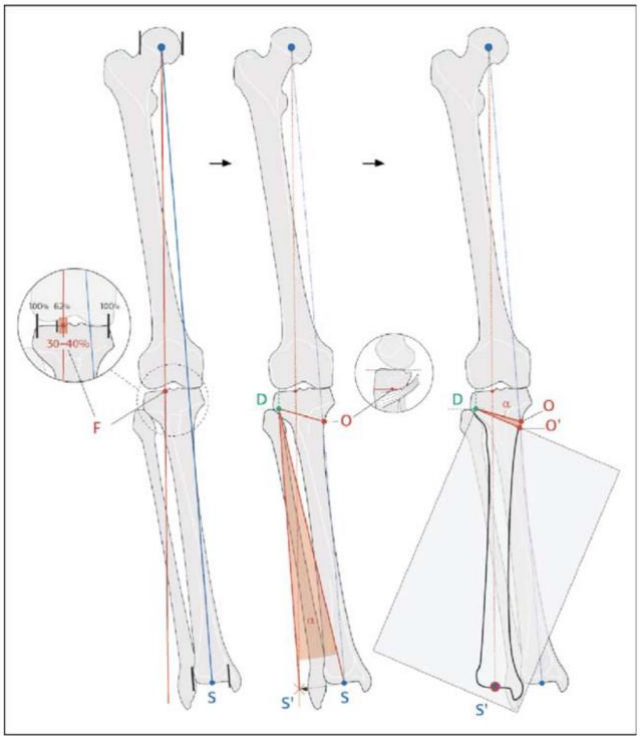


Fig. 6: Preoperative planning of an HTO surgery with an overcorrection of the new mechanical axis
reprinted from Staubli et al., (2003)

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The underlying rationale of HTO is the modification of load distribution within the knee joint. In cases of medial compartment overload, the procedure aims to reduce medial joint loading by redistributing the overall compressive forces towards the lateral compartment (De Pieri et al., 2023; Van Egmond et al., 2017). This redistribution is accompanied by a reduction in the KAM, with the magnitude of KAM decrease being proportional to the degree of alignment correction achieved (Van Egmond et al., 2017). Consequently, postoperative biomechanical gait characteristics have been reported to approximate those of healthy individuals, indicating a restoration of more physiological loading patterns (Van Egmond et al., 2017).

1.3.2 Conservative Treatment

Individuals with anterior cruciate ligament reconstruction (ACLR) represent a population particularly susceptible to the development of early PTOA of the knee joint, especially within the medial compartment (Nilsson et al., 2025). This is because of interlimb (Murrell-Smith et al., 2025) and intralimb (Sigward et al., 2018) compensation strategies as well as altered joint mechanics, including increased anterior tibial translation, external tibial rotation, and shifts in load-bearing regions during gait (Murrell-Smith et al., 2025; Pandy et al., 2025), in addition to malalignment of the lower limb (Nilsson et al., 2025). These alterations may contribute to abnormal cartilage loading patterns and mechanical stress distribution within the knee joint, thereby promoting the development and progression of PTOA following ACLR (Pandy et al., 2025). This underscores the importance of targeted rehabilitation strategies by means of identifying and addressing such maladaptive patterns.

For the case of OA being already present, recent evidence-based guidelines emphasise that the conservative management of knee OA should consist of a multimodal approach, as no single intervention sufficiently addresses the whole spectrum of clinical outcomes associated with the disease (Allaeys et al., 2020).

Within this framework, interventions with the strongest evidence for beneficial effects include weight reduction in individuals with a body mass index exceeding 28 kg/m², structured exercise therapy — including aerobic, strengthening, and proprioceptive components — as well as selected pharmacological treatments such as non-steroidal anti-inflammatory drugs (NSAIDs) and intra-articular injections of corticosteroids and hyaluronic acid (Allaeys et al., 2020).

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A central pillar of this multimodal approach is physical exercise therapy. Current recommendations highlight that exercise programmes should be heterogeneous, integrating functional exercises, neuromuscular and proprioceptive training, strength training, and aerobic conditioning (Allaeys et al., 2020). One specific exercises for instance might be the FL (Murrell-Smith et al., 2025) whose advantage it is to be a versatile and highly adaptable exercise (R. F. Escamilla et al., 2025).

Within strength training interventions, particular attention has been given to the role of hip musculature, especially the hip abductors. Evidence indicates that hip abductor strengthening programs, carried out in addition to general lower-limb-targeting strengthening programs, lead to significant acceleration in relieve of symptoms like pain as well as in faster improvements in functional outcomes (Yuenyongviwat et al., 2020).

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1.4 Current Evidence

A variety of recent studies have been listed in the below Table 1 to give a comprehensive representation of the diverse approaches to examine the FL exercise by means of analysis of specific biomechanical variables.

Table 1: Recent scientific publications of experimental studies which biomechanically examined the FL (provided information is regarding the knee joint, unless marked otherwise)

Source	Objectives	Participants	Results
Widhalm et al., (2025) [a]	Influence of physical activity level on frontal knee alignment	64 (55 % female) healthy exercised as well as untrained	Physical activity levels don't affect frontal knee angles
Gao et al., (2025)	Influence of different types of weight bearings and training experience in kinematics, kinetics, muscle activations	24 (0 % female) fitness trainees (different levels of experience)	Training-experience and weight-bearing-type both had a significant influence in kinematics and kinetics [d]
Escamilla et al, (2025)	Influence of step-length on muscle recruitment patterns	2o (50 % female) healthy	Step-length had an influence in muscle recruitment [e]
Gao et al., (2023) [b]	Influence of targeted GMED fatigue on kinematics, kinetics, muscle activations	15 (0 % female) healthy	Targeted GMED fatigue had an influence in kinematics, kinetics and muscle activations [f]
Bezerra, (2021)	Influence of trunk inclination and step-variation on muscle activation	12 (100 % female) healthy	Trunk inclination and step-variation both had an influence in muscle activation [g]
Alkjær et al., (2020) [c]	Influence of ACL-Injury on kinematics, kinetics, muscle activations	28 (39 % female) ACL-injured, 28 (39 % female) healthy controls	Significant differences in all examined metrics have been observed [h]

[a] Exercise of split lunge has been observed (no forward stepping within then movement)

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[b] Observed net-joint-moment regarding kinetics solely, no KAM

[c] ACL-injured patients underwent ACLR and were 10 months post-surgery at the timepoint of assessment

[d] Hip, knee, and ankle ROM and peak-angles were significantly greater in novices than in seniors, dumbbells reduced hip moments and stiffness in novices, barbells increased ankle moments in seniors

[e] GMED activity was significantly higher with higher step-length

[f] sagittal and horizontal planes of the hip and knee joints, in both, the peak angles and joint-ROM increased, the net-joint-moment in the sagittal plane of the knee joint decreased, quadriceps mean and maximum force significantly increased

[g] Higher activation in the erector spinae has been observed when trunk was inclined as well as during the walking-lunge-variation, higher activation in the gluteus maximus was observed during step-forward-variation and walking-lunge-variation

[h] Significantly lower peak-knee-flexion, peak-knee-angular-velocity, peak-knee-net-joint-moment and mean-GMED-amplitude

As stated in the above Table 1, recent experimental investigations on the FL have increasingly examined the influence of movement execution, loading strategy, fatigue, and pathological conditions on lower-limb biomechanics, highlighting the growing scientific interest in this functional task.

Training experience and weight-bearing modality were shown to significantly influence lower-limb kinematics and kinetics during the FL (Gao et al., 2025), while variations in step length and trunk inclination affected muscle recruitment patterns, particularly activation of the gluteal and trunk musculature (Bezerra, 2021; R. F. Escamilla et al., 2025). Furthermore, targeted GMED fatigue altered hip and knee kinematics, joint ROM, Knee-Net-Joint-Moments, and quadriceps muscle forces (Gao et al., 2023). Altered biomechanical patterns have also been observed in pathological populations, as individuals following ACLR demonstrated reduced knee flexion, knee angular velocity, Knee-Net-Joint-Moments, and GMED activation compared to healthy controls (Alkjær et al., 2020).

Collectively, these findings indicate that FL biomechanics are highly sensitive to neuromuscular function, movement strategy, and pathological status. However, despite the growing body of literature, substantial gaps remain regarding frontal-plane knee loading during FL execution. In particular, most studies have focused on sagittal kinematic

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and kinetic variables, whereas the role of the KAM and compartment-specific knee loading remains insufficiently investigated.

The below listed Table 2 gives a widespread overview of recent scientific publications which examined compensation mechanisms and further biomechanical effects after application of different methods of weakening or strengthening the GMED muscle.

Table 2: Recent scientific publications of experimental studies which biomechanically examined the compensation mechanisms and effects of a variety of methodological inductions of GMED fatigue or strengthening

Source	Motion	Method	Results
Kothurkar et al., (2025)	Squat	Reduction of GMED-strength within computational model	Compensation of reduced GMED strength by increased activation of thigh muscles followed by increased KJCF [a]
Yamagata et al., (2024)	Stair ascent and descent	Reduction of GMED-strength within computational model	Increase of medial KJCF, decrease of lateral KJCF
Dimitriou et al., (2024)	Squat	Superior gluteal nerve block	No differences in kinematics, decrease in ipsilateral JRF and increase in contralateral JRF of the knee
Jolas et al., (2023)	Walking	Increase of GMED-strength within computational model	First peak KJCF was reduced and shifted to GMED via activation of hamstrings
Yamagata et al., (2022)	Walking	Reduction of GMED-strength within computational model	Increase of medial KJCF, decrease of lateral KJCF
Tang et al., (2022)	Walking	Reduction of GMED-strength by fatigue-protocol	1st and 2nd peak of KAM were increased compared to the non-fatigue-condition

[a] Knee joint contact force (KJCF)

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A scope in current scientific literature recognises the GMED as an important contributor to lower-limb biomechanics and knee joint loading. Various methodological approaches have been used to investigate the biomechanical consequences of altered GMED function, including computational modelling, experimentally induced fatigue, and iatrogenic weakening. Computational studies demonstrated that reductions in GMED strength increase medial KJCF and decrease lateral KJCF during walking and stair ambulation (Yamagata et al., 2022, 2024). Similarly, simulated GMED weakness during squatting resulted in compensatory increases in thigh muscle activation, accompanied by elevated KJCF (Kothurkar et al., 2025). In contrast, computational strengthening of the GMED during walking reduced the first peak of KJCF and redistributed loading demands through altered hamstring activation patterns (Jolas et al., 2023).

Beyond simulation-based approaches, experimentally induced GMED fatigue has likewise been associated with altered frontal-plane knee loading, as reflected by increases in both the first and second peaks of the KAM during walking (Tang et al., 2022). Furthermore, iatrogenic weakening induced by superior gluteal nerve block resulted in altered joint reaction force distribution between limbs during squatting, despite unchanged movement kinematics (Dimitriou et al., 2024).

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1.5 Thesis Goals

The KAM represents a key biomechanical parameter associated with intercondylar knee joint loading distribution and the development and progression of OA (D'Souza et al., 2022). Emerging evidence highlights a strong connection between KAM and the function of the GMED, indicating a direct neuromuscular influence on knee joint loading (Tang et al., 2022).

This relationship is of particular clinical relevance in populations such as individuals following ACLR, who are at increased risk of developing PTOA of the knee and exhibit persistent compensatory movement strategies, including altered joint kinematics and interlimb asymmetries (Murrell-Smith et al., 2025). Despite the recognised importance of the KAM, the latter remains insufficiently investigated in common rehabilitation exercises, limiting their integration into evidence-based clinical decision-making (Murrell-Smith et al., 2025).

The generation of normative KAM data in healthy individuals is therefore essential to establish reference values, enabling meaningful comparisons with pathological populations and the establishment of safe progression criteria during rehabilitation (Murrell-Smith et al., 2025).

In this context, the FL exercise represents a fundamental yet underexplored rehabilitation exercise, widely used in ACLR recovery to promote strength and functional stability. However, no data currently exists regarding the KAM during the execution of the FL. This thesis therefore aims to address this gap by providing biomechanical data on KAM in healthy individuals during the FL, thereby contributing to the evidence base for clinical assessment and rehabilitation planning. Furthermore, the influence of hip abductor muscle fatigue on the KAM during the FL is explored.

Primary Goals of this thesis:

1) Quantification of normative data of healthy male adults for the KAM of the dominant limb during the FL exercise

2) Investigation of the influence of targeted fatigue of the GMED of the dominant limb on the KAM during FL exercise

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2 Materials & Methods

The present study aimed to follow current state-of-the-art methodology in biomechanical movement analysis by applying three-dimensional motion capture in combination with musculoskeletal modelling. This approach has increasingly become the standard for estimating internal joint loads and dynamic outcome parameters, particularly in the investigation of knee mechanics within biomechanical research (Mündermann et al., 2024).

2.1 Design

A single-group experimental pre–post design was conducted in the motion analysis laboratory at TU-Wien, with all measurements acquired within a single session per participant. The methodological framework of the present study was closely based on the protocol described by (Gao et al., 2023), particularly regarding the investigated movement task and the sequential laboratory procedure. However, the present study differed in its primary biomechanical outcome measure, with a specific focus on the KAM. Efforts were made to adhere to current state-of-the-art biomechanical methodology throughout the experimental workflow. All experimental procedures were conducted by investigators with substantial expertise in biomechanics and clinical practice. The study population consisted of young, healthy adults, all of whom provided written informed consent prior to participation.

2.1.1 Participants

Ten male young healthy participants were recruited for the present study. The mean age of the participants was 28.9 years ± 3.6 years, with a body mass of 81.7 kg ± 11.5 kg, a height of 182.2 cm ± 4.9 cm and a BMI of 24.5 kg/m^2± 2.5 kg/m^2.

The level of physical activity was assessed with the European Health Interview Survey – Physical Activity Questionnaire (EHIS-PAQ) (Finger et al., 2015) (Appendix A).

Inclusion-, and exclusion criteria which must be met for voluntary participation are listed in the below-mentioned Table 3.

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Table 3: Inclusion-, and exclusion criteria for eligible participants

Inclusion criteria	Exclusion criteria
age between 18 and 38 a	acute musculoskeletal pain at rest or during required motor task
BMI between 20 and 28 kg/m²	professional or elite athlete
good perceived general health condition	history of major lower-extremity injury with or without surgical intervention
present in rested condition when attending lab-session	incapability to perform the motor task for any reason
provision of written informed consent	

2.1.2 Materials

Surface Electromyography (sEMG)

sEMG signals were recorded using Cortex motion capture software (Motion Analysis Corporation, 2001) via the Trigno™ Wireless EMG System with wireless EMG sensors (Delsys Inc., Boston, MA, USA) in order to record activation patterns of highly relevant muscles for the performance of the FL. Signals of all sensors were simultaneously recorded at a sampling frequency of 1.2 kHz. A complete overview of all instrumented muscles and sensor locations is provided in Table 17 and Fig. 7 of this thesis.

Force-Plates

Three force plates (Manufacturer: Advanced Mechanical Technology Inc., Model BMS400600-2K), integrated into the laboratory floor, were used to acquire kinetic data during the experiment. The force plates recorded ground reaction forces and ground reaction moments in all three spatial directions at a sampling frequency of 1.2 kHz which allowed for the calculation of the centre of pressure which has been used for subsequent post-processing. The setup consisted of one rear force plate and two front force plates.

Resistance-Bands

Resistance loop bands (ZenLoops Fabric 5-piece set, Zen One Sports, Germany) were used as part of the exercise setup. According to the manufacturer, the set comprises five resistance levels ranging from “X-light” to “X-heavy” (Zen One Sports, 2026). Details to the used material are mentioned in Table 24 in the appendix of this thesis.

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Motion-Capture-System

Fifteen infrared cameras (Manufacturer: Motion Analysis.com, Model: Kestrel) were used to capture the positions of passive reflective spherical markers (Qualisys Motion Capture Systems, diameter 13 mm) within the laboratory. The cameras were arranged in a rectangular configuration and mounted in elevated positions near the ceiling, providing full coverage of the capture volume, which was sufficient to encompass all participant's movements during the FL. Marker positions were recorded at 120 Hz using Cortex motion capture software (Motion Analysis Corporation, 2001), while force plate data sampling was synchronised with the optical motion capture system. As a representative verification of system performance, a known inter-marker distance of 500 mm was repeatedly measured. The optical motion capture system yielded a mean measured distance of 499.67 mm (SD = 0.54 mm), demonstrating close agreement with the reference value and low measurement variability.

EHIS-PAQ

The EHIS-PAQ is a self-reported questionnaire (Finger et al., 2015) recommended for population-level assessment of physical activity. It consists of eight questions covering the domains of work, transport, leisure-time, sedentary behaviour and sport-activities while distinguishing between aerobic and muscle-strengthening activities (Sember et al., 2020). The detailed questionnaire including its evaluation is mentioned under Appendix A of this thesis.

2.2 Laboratory - Procedure

This chapter accounts for the chronology of each of the laboratory sessions in which the subjects have been assessed.

2.2.1 Materials Preparation

Before each laboratory-session the required equipment was prepared appropriately. Motion capture system, including force-plates, cameras and computer with capture software were booted as well as force-plates were calibrated to zero. Light conditions were properly adapted by shutting window-sheets and removal of any reflective surface which might have an influence in marker-detection by the camera system. All objects which might interfere with any camera-field have been removed. Then the motion capture system was calibrated using a standardised L-shaped frame to define the global laboratory coordinate

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system. In addition, a standardised calibration wand was moved throughout the capture volume to calibrate the camera arrangement and verify accurate three-dimensional marker reconstruction across the measurement area. Finally, all sEMG-sensors have been switched on and checked for functionality.

2.2.2 Introduction of the Participant

Upon arrival at the laboratory, participants were first familiarised with the facilities and received a detailed explanation of the chronological sequence of the upcoming experimental procedure. They were provided with contact details of the responsible study personnel, including the investigators and examiners, and were informed that questions could be asked at any time throughout the study. Furthermore, participants were explicitly informed of their right to withdraw from the experiment at any point without disadvantage and without the need to provide a reason.

Thereafter, participants were given sufficient time to read and sign the informed consent form, which addressed both the voluntary nature of participation and the processing of personal data. Finally, participants were provided with adequate time to complete the EHIS-PAQ questionnaire (Finger et al., 2015).

2.2.3 Application of Markers and Sensors

After changing into appropriate clothing, participants were prepared for instrumentation with surface electromyography (sEMG) sensors and optical reflective markers.

In a first step, sEMG sensors were attached to predefined muscle sites in accordance with the recommendations of the SENIAM project (Surface ElectroMyoGraphy for the Non-Invasive Assessment of Muscles) (Seniam.Org, 2026). The complete list of monitored muscles is provided in the appendix of this thesis. Prior to sensor placement, the skin was prepared using a skin-compatible isopropanol solution to reduce skin impedance and optimise signal quality at the respective electrode sites. Special attention was paid to accurate sensor placement and alignment with the primary direction of the underlying muscle fibres as well as the prevention of potential movement artifacts caused by interaction with clothing during movement. After application, sensor functionality was verified and signal quality was assessed based on the response to voluntary muscle activation.

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Following sEMG preparation, reflective markers were placed on predefined anatomical landmarks and tracking locations required for motion capture. An example of a fully instrumented participant is presented in Fig. 7. A detailed overview of all anatomical and tracking marker locations is provided in Table 20 in the appendix.

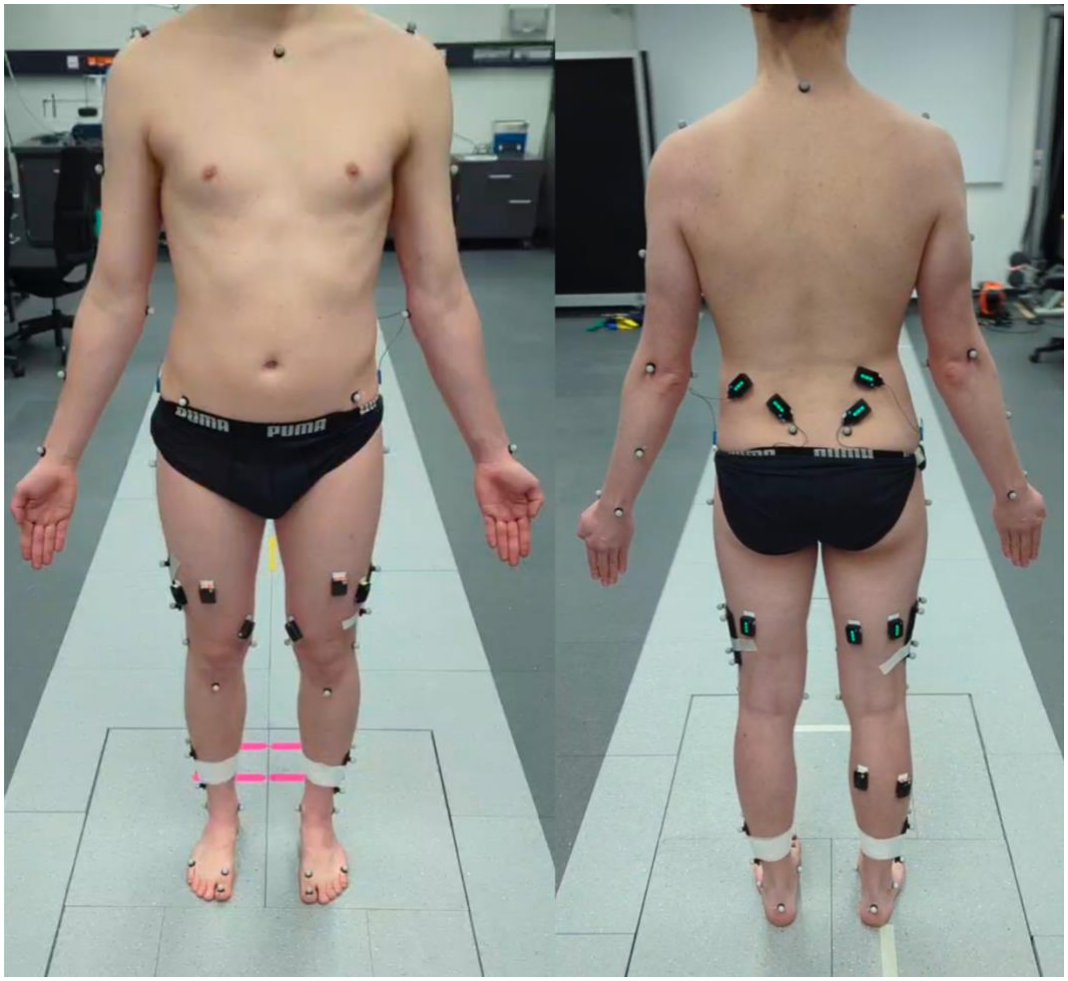


Fig. 7: Experimental setup showing a participant equipped with an optical marker set and sEMG sensors standing on force plates in the anatomical position (functionally dominant side: right)

2.2.4 Determination of functional leg dominance and stride length

As the primary aim of this study was to investigate biomechanical outcome measures of the dominant leg, particularly the KAM, the dominant leg was defined as the front leg during the FL. To determine leg dominance, a push test was conducted by one of the examiners. Prior to testing, participants received a brief explanation of the procedure and were informed that a manual perturbation would be applied, ensuring that the push itself was anticipated while the exact timing remained unknown.

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For this purpose, the participant stood upright in a relaxed bipedal stance while the examiner positioned himself behind the participant and applied a sudden, unexpected forward perturbation. In response, the participant performed a compensatory forward step with one leg to maintain balance and avoid falling. The leg used for the initial recovery step was classified as the dominant leg for the purposes of this study.

To ensure standardised execution of the FL task and improve comparability across participants, step length was determined relative to each participant's leg length. Therefore, lower-extremity and pelvic anthropometric measurements were obtained using a tape measure. Specifically, the vertical distance from the anterior superior iliac spine (ASIS) to the ground as well as the inter-ASIS distance were measured. These parameters were used in an adapted application of the hip joint centre regression equations described by Harrington et al. (2007) to estimate the vertical distance between the ASIS and the hip joint centre. This estimated value was subsequently subtracted from the measured ASIS-to-ground distance to approximate leg length. Based on the resulting estimate, the target stride length for the FL was calculated and marked on the laboratory floor using adhesive tape, thereby providing participants with a visual guide for movement execution (Fig. 9).

2.2.5 Experiment

After completion of the previously described preparatory steps, including participant briefing and instrumentation, the experimental procedure could be started. An overview of the chronological sequence and general structure of the laboratory session is provided in Fig. 8.

Prior to data collection, both the FL and the imposed fatigue exercise were demonstrated by the examiner. Participants then performed a standardised warm-up consisting of ten bodyweight squats and ten jumping jacks. Subsequently, participants were given the opportunity to familiarise themselves with the experimental tasks by completing one practice set of the FL and one practice set of the fatigue exercise. A rest period of more than five minutes was provided between familiarisation and the beginning of the recorded trials.

The central component of the laboratory session consisted of motion capture recordings of one pre-fatigue trial of the FL, followed by the imposed fatigue exercise, and one post-fatigue trial of the FL. Each trial comprised ten repetitions performed with the functionally dominant leg as the front leg, immediately followed by ten repetitions performed with the

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non-dominant leg as the front leg, without a rest period between sides. Execution of the FL was standardised using acoustic pacing provided by a metronome.

Immediately after completion of the pre-fatigue motion capture trial, participants transitioned without relevant delay to the fatigue protocol. The fatigue protocol is geared to the work of Rabello et al. (2022) and consisted of four sets performed up to task failure, with 90 seconds of rest between sets. Each repetition was standardised to a duration of four seconds, comprising a two-second concentric phase and a two-second eccentric phase, with pacing controlled by a metronome. The initial resistance was selected by the examiner such that participants were expected to achieve a repetition count between eight and fifteen while maintaining the prescribed movement rhythm, full range of motion, and appropriate movement quality. A set was terminated either by the examiner, in the presence of relevant movement deterioration, or by the participant once a rating of perceived exertion of at least 19 on a 20-point Borg scale (Borg, 1998) was reached. Relevant movement deterioration included failure to maintain the imposed rhythm, loss of full range of motion, or the occurrence of compensatory movement strategies, such as excessive pelvic displacement, lateral trunk flexion, or an excessively forceful initiation of the movement. Depending on the number of completed repetitions, the resistance was adjusted from set to set: if eight or fewer repetitions were completed, resistance was decreased for the subsequent set, whereas if fifteen or more repetitions were completed, resistance was increased. Throughout the fatigue protocol, the same examiner provided continuous feedback regarding movement quality as well as standardised verbal encouragement.

Following completion of the fourth set of the fatigue protocol, care was taken to minimise the transition time between the end of the final fatigue set and the start of the post-fatigue motion capture trial. Transition time was defined as the time interval between termination of the fourth fatigue set and initiation of the first post-fatigue FL repetition. This interval was recorded for each participant and is reported and discussed in later sections of this thesis.

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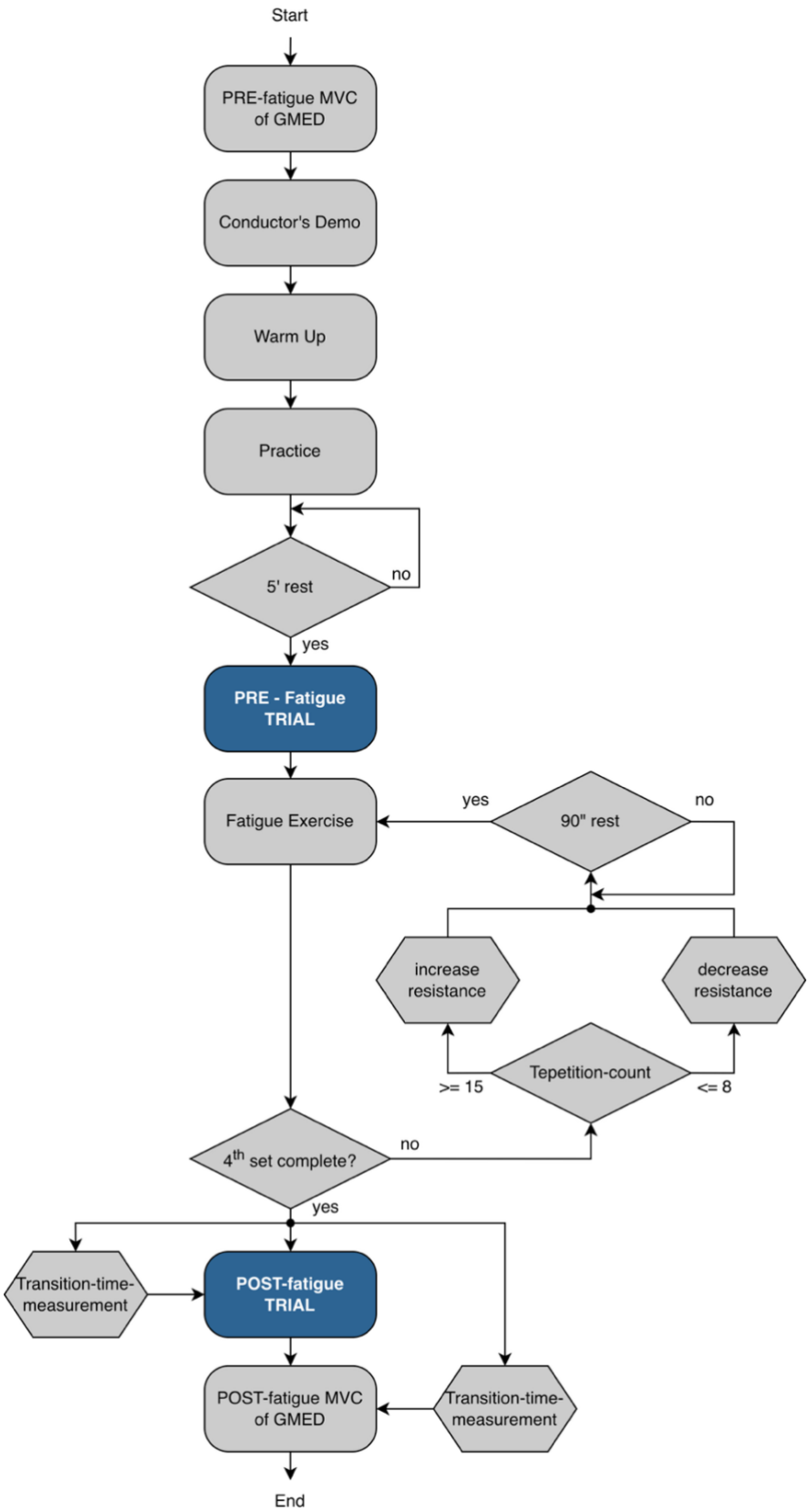


Fig. 8: Overview of the experimental procedure showing the sequence of MVIC measurements, participant familiarisation with the FL, fatigue exercise and pre- and post-fatigue motion capture of the FL

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2.2.5.1 Forward Lunge Exercise (FL)

One repetition of the FL variation considered in this thesis started from an upright bipedal standing position with both feet placed on the rear force-plate (Fig. 9a). From this starting position, one foot was lifted and moved anteriorly to perform a forward step onto the respective front force-plate, while the distal aspect of contralateral limb's foot remained in contact with the rear force-plate (Fig. 9b). The movement continued until the participant reached the lowest position of the exercise, defined by the rear knee descending to approximately ankle height (Fig. 9c). In this position the trunk has been maintained in a reasonably upright posture. Subsequently, the movement was reversed (Fig. 9d and Fig. 9e), and the participant returned to the initial upright bipedal standing position on the rear force-plate, thereby completing one repetition (Fig. 9f).

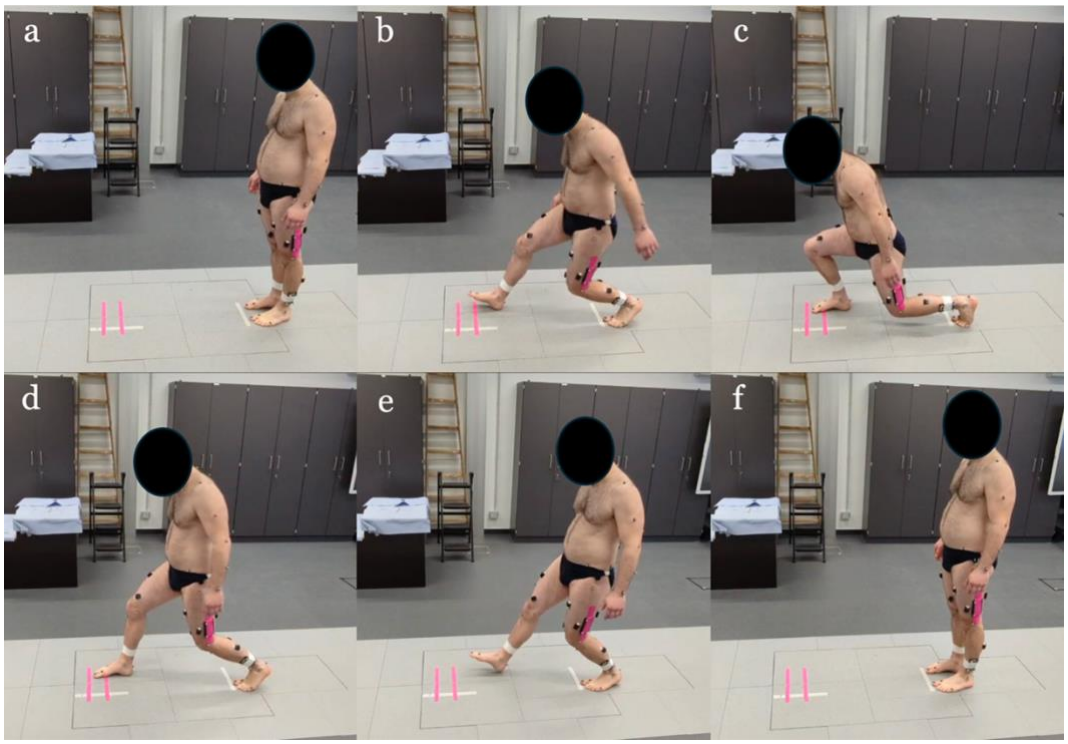


Fig. 9: Participant performing the FL in the laboratory

a) Starting-Position, b) Initial Contact with the Front Force-Plate, c) Lowest Position

d) Approximately 50 % of Concentric Phase, e) Time-Point of "Heel-Off", f) Final Position

2.2.5.2 Fatigue Exercise (FE)

For the FE, participants were provided with the resistance loop described in Section 2.1.2 Materials, which was positioned around both forefeet. The exercise consisted of repeated standing hip abduction movements performed against the resistance of the loop

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(Fig. 10a-d). During each repetition, one leg served as the moving limb, while the contralateral limb remained in contact with the ground as the stance limb. The end position of each repetition was defined by contact between the lateral aspect of the moving foot and the wall. The distance between the functional axis of the stance foot and the wall was set to 52 cm, resulting in approximately 30° of hip abduction across participants when reaching the wall-contact position. To ensure balance, participants were permitted to hold onto a stable support throughout execution of the exercise.

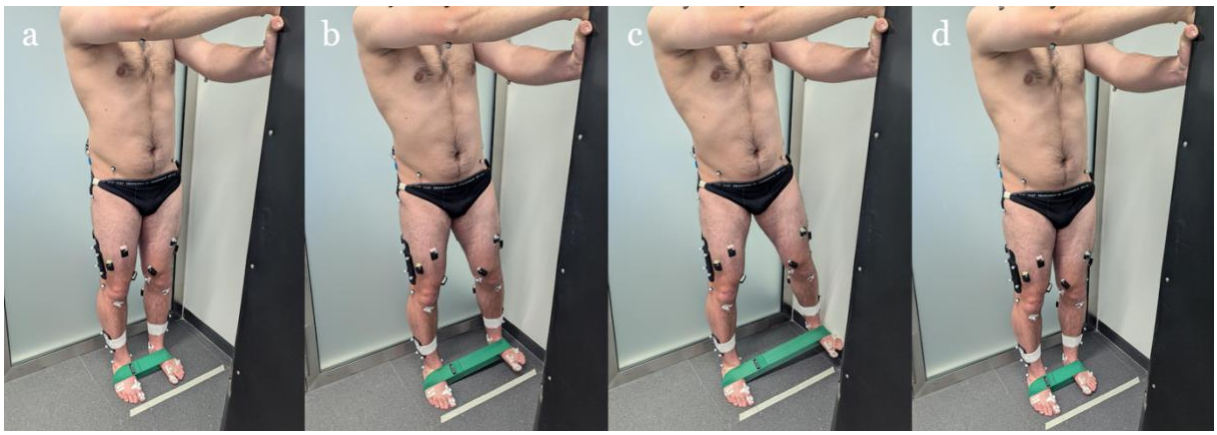


Fig. 10: Participant performing the fatigue exercise
a) Starting-Position, b) Approximately 50 % of Concentric Phase,
c) 100 % of Concentric Phase (Touching of the Wall), d) Final-Position

2.2.6 Maximum voluntary isometric contraction (MVIC)

MVICs were recorded for a group of muscles which has been chosen after careful consideration of current scientific literature examining muscle activities during lower-limb functional exercises. In the current thesis, only the sEMG of the dominant GMED has been evaluated. A description of the specific testing position of the GMED is retrievable from Table 4 with an exemplary photograph of the MVIC testing position viewable under Fig. 11. Corresponding information for the remaining muscles instrumented is listed in the appendix of this thesis.

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Table 4: Testing-Position of the GMED during MVIC

Muscle	Testing-Position	
GMED	participant's position	side-lying
	hip-rotation	0 °
	hip-abduction	10 °
	knee-flexion	0 °

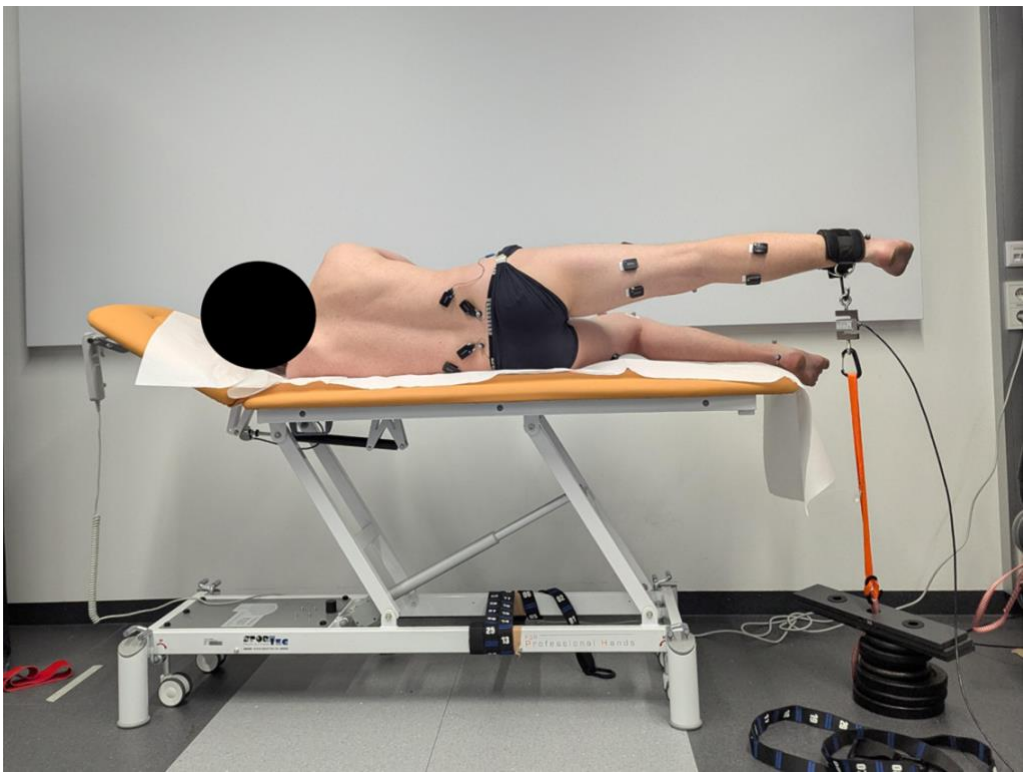


Fig. 11: Participant positioned for GMED MVIC testing

To generate external resistance, a non-extensible strap was placed around the relevant limb segment and mounted to a stable external fixation point. Prior to each recorded contraction, participants were instructed to gradually remove any slack from the strap during a standardised three-second countdown. Immediately thereafter, an acoustic signal was provided by the examiner, upon which the participant was instructed to exert a maximal isometric contraction and maintain this effort for five seconds. Two MVIC trials were recorded for every muscle. A rest interval of one minute was provided between the two trials of the same muscle. In addition, more than one minute of rest was generally available when changing between testing positions and reconfiguring the strap fixation for the subsequent muscle.

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Throughout all MVIC trials, participants were instructed to maintain the prescribed testing posture and avoid compensatory movements as far as possible. To ensure consistent execution, all participants were permitted to hold onto the treatment bench throughout all MVIC trials if preferred. All MVIC trials were supervised by the same examiner, who was responsible for participant positioning, acoustic cueing, and provision of standardised verbal encouragement.

2.3 Post-Processing – Procedure

The post-processing procedure comprised a multi-step computational workflow that transformed the recorded raw motion capture and ground reaction force (GRF) data, in combination with an unscaled generic musculoskeletal model and participant-specific anthropometric information, into the primary biomechanical outcome of interest, namely the KAM. This workflow consisted of several consecutive processing stages, each of which contributed to the derivation of the final outcome variable. A schematic overview of the overall procedure is provided in Fig. 12. The individual sub-processes and their methodological implementation are described in detail in the following sections of this chapter.

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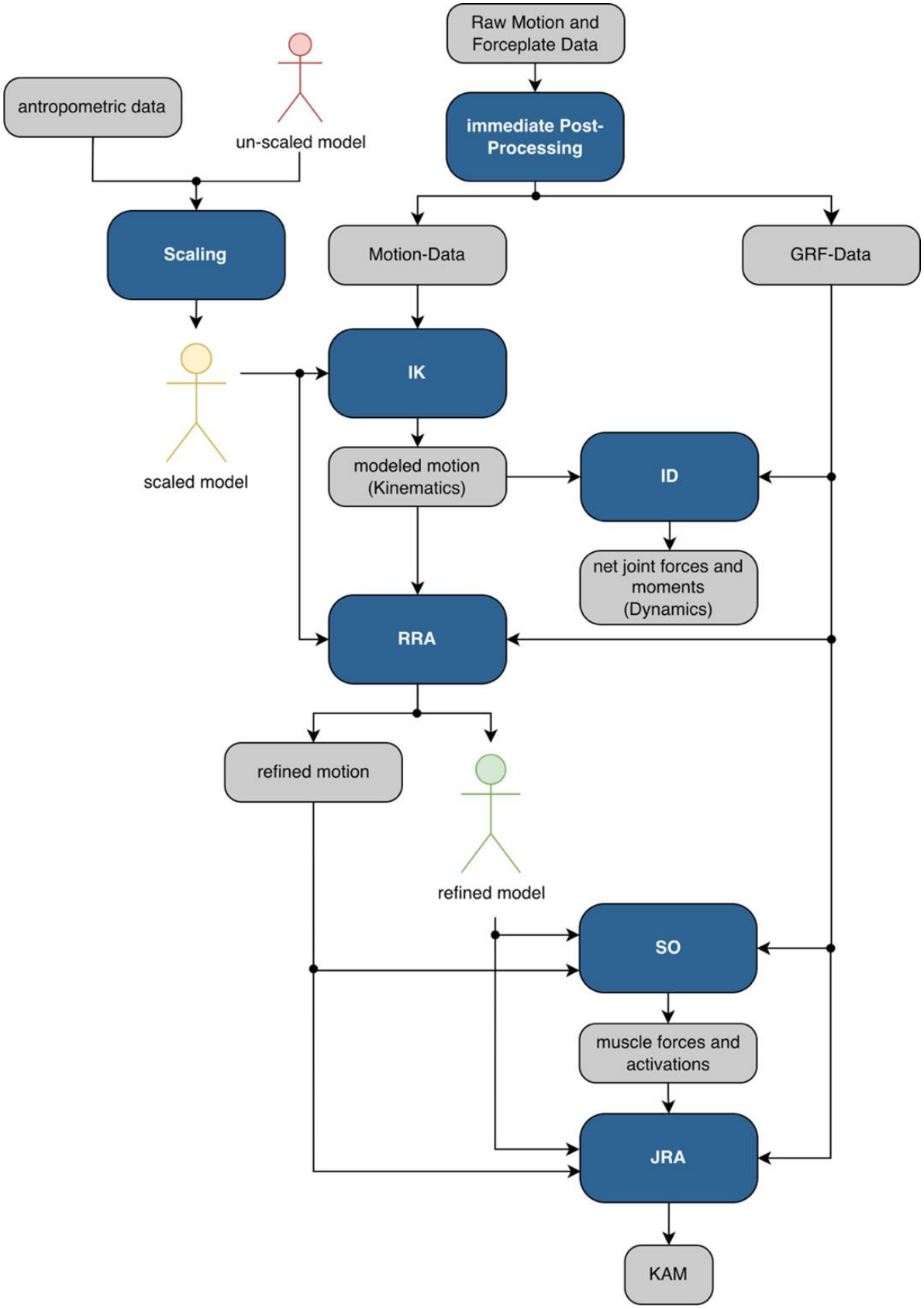


Fig. 12: Flowchart of the post-processing workflow used to estimate the KAM from motion-capture, force-plate and anthropometric data

Blue blocks indicate data-processing steps.

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2.3.1 Immediate post-processing

To obtain a standardised and processable *.c3d*-file from the raw camera data recorded by the motion capture system, an initial post-processing procedure was performed using *Cortex* (Motion Analysis Corporation, 2001) in the version of 10.4.0.47 (64 bit version). This software enables real-time motion capture, three-dimensional reconstruction of marker trajectories, marker labelling, and gap filling, thereby providing a consistent dataset for subsequent kinematic post-processing.

During this initial post-processing stage, individual marker trajectories were visually inspected, and artefacts or trajectory distortions were corrected using the integrated processing tools available in Cortex, including *Cubic Join* and *Virtual Join*. In addition, occasional marker swaps resulting from automated misidentification by the software were identified and corrected manually where necessary. This procedure resulted in one consistent *.c3d*-file for each trial (pre- and post-exhaustion), containing synchronised ground GRF-data and three-dimensional kinematic marker trajectories, as well as not-yet temporarily aligned sEMG recordings.

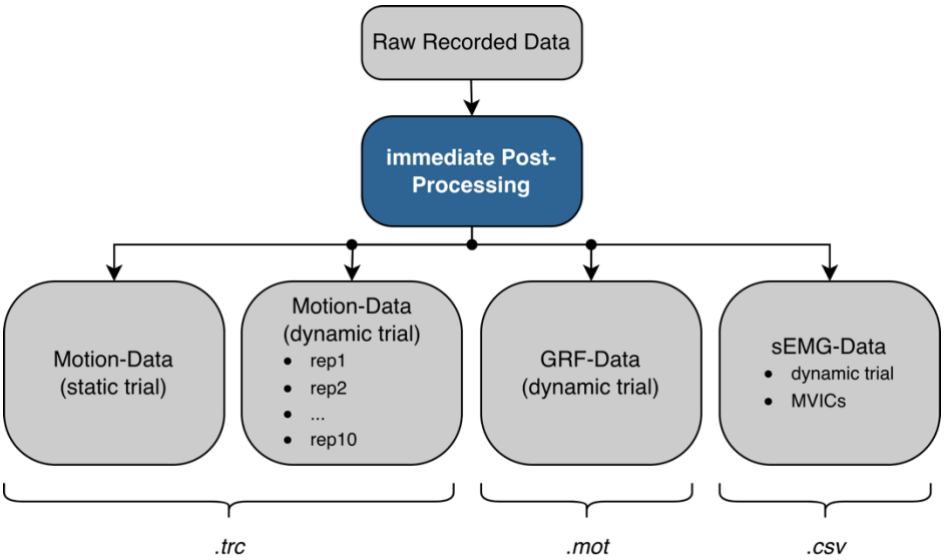


Fig. 13: Inputs and outputs of the immediate post-processing step, transforming raw experimental recordings into motion, GRF, and sEMG datasets

In a subsequent processing step, *Python* (Python Software Foundation, 2025) was used for marker set handling and preprocessing. This included the computation of virtual markers required for subsequent model scaling, such as estimated joint centre markers

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for the hip, knee, and ankle, as well as auxiliary pelvic reference markers including the midpoint between the anterior superior iliac spines (mid-ASIS) and the midpoint between the posterior superior iliac spines (mid-PSIS). Hip joint centres were estimated using the regression equations proposed by Hara et al. (2016), whereas knee and ankle joint centres were determined as the arithmetic mean of the corresponding medial and lateral femoral epicondyle markers and medial and lateral malleolus markers, respectively. In addition, virtual foot markers were generated by projecting the experimentally recorded foot markers onto ground level. These derived markers were used to define anatomically and functionally relevant reference points for the determination of scaling factors in the subsequent model scaling procedure.

Python further has been used in another consecutive step, in which the data stored within in each *.c3d*—file got extracted into separate file formats for further processing. Specifically, marker trajectory data were exported as *.trc*-files, GRF data as *.mot*—files. Each of these files served as input for the subsequent post-processing pipeline (Fig. 13).

Regarding the objectives of this thesis, the relevant temporal window of each repetition of the FL was defined as the period during which the striking foot was in contact with the ground. Accordingly, the kinematic data were segmented into individual files, each representing a single repetition-specific ground-contact event. Ground contact onset was defined as the first time point at which the vertical component of the GRF exceeded a threshold of 50 N. Ground contact offset was defined as the final time point within the corresponding GRF burst at which the vertical GRF remained above 500 N. To reduce the influence of transient artefacts, only GRF bursts exceeding a minimum duration threshold of 0.5 s were considered valid events. Finally, each segmented repetition was visually inspected for plausibility using the graphical user interface of *OpenSim*.

2.3.2 OpenSim Tools

This section is dedicated to the setup of the post-processing pipeline established as consecutive processing steps of integrated tools of the software of OpenSim.

2.3.2.1 Scaling

The formerly introduced musculoskeletal model *RajagopalLaiUhlrich2023* (Lai et al., 2017; Rajagopal et al., 2016; Uhlrich et al., 2022) (chapter 1.2.4.1 Scaling) has been completed using a marker-set comprising of 76 markers, including 56 experimental

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markers including 16 tracking markers and 20 virtual markers, which has thoughtfully been chosen to optimally image the physical movement of the subject. Thereby, joint-centre-markers for the knee and ankle have been calculated by the arithmetic mean of medial and lateral epicondyle markers, and medial and lateral malleoli markers respectively. Joint-centre-markers for the hip have been calculated by using regression formulae provided by Hara et al. (2016). Below-mentioned Fig. 14 depicts the unscaled model and the marker-set within the OpenSim graphical user interface. Furthermore, tables listing the marker-set and the scale-factors as well as marker-weightings used for the scaling process are provided in Table 20, Table 22, Table 21 in the appendix of this thesis.

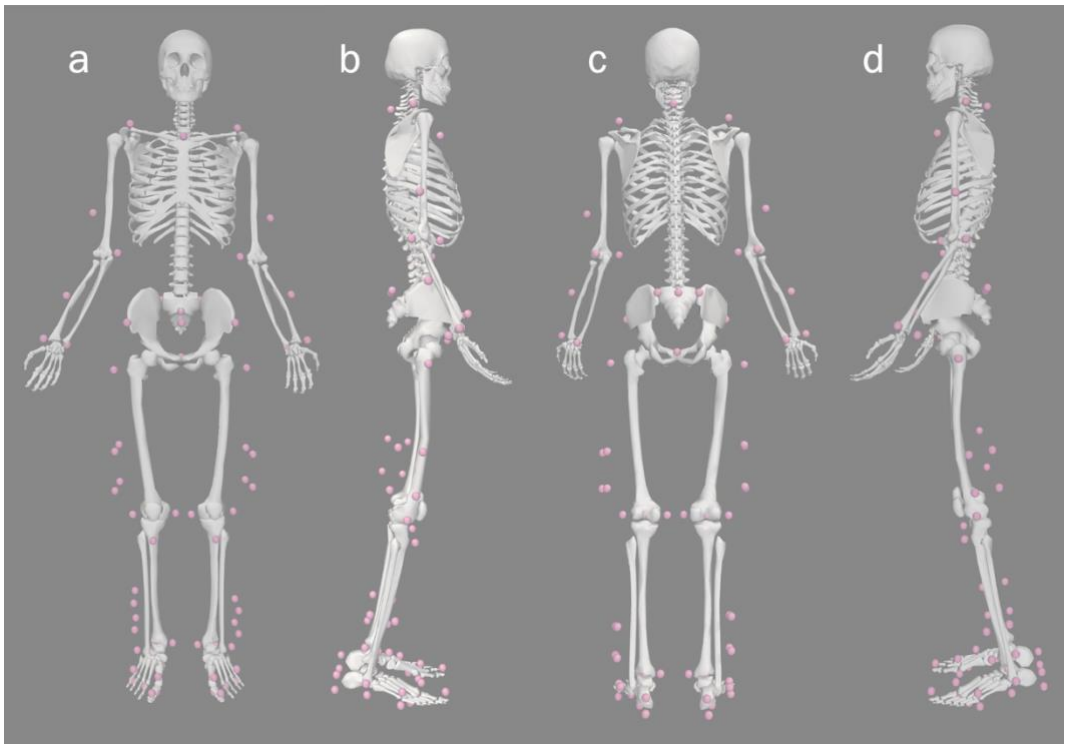


Fig. 14: Unscaled generic *RajagopalLaiUhlrich2023* (Lai et al., 2017; Rajagopal et al., 2016; Uhlrich et al., 2022) musculoskeletal model with full experimental marker set displayed in OpenSim's GUI, shown in anatomical position across (a) frontal, (b) right side, (c) posterior, (d) left side views (muscles omitted for clarity)

For the actual process of scaling (Fig. 15) the Automatic Scaling Tool (AST) of Di Pietro et al. (2024) has been used. It automates the process by iteratively adjusting virtual markers of the generic model using the provided scaling determinants and the kinematics of the static trial. The AST thereby evaluates the root mean square (RMS) error and maximum

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marker error, implementing corrective actions until achieving the desired level of accuracy of an upper threshold of 4 mm for the maximum marker error is met.

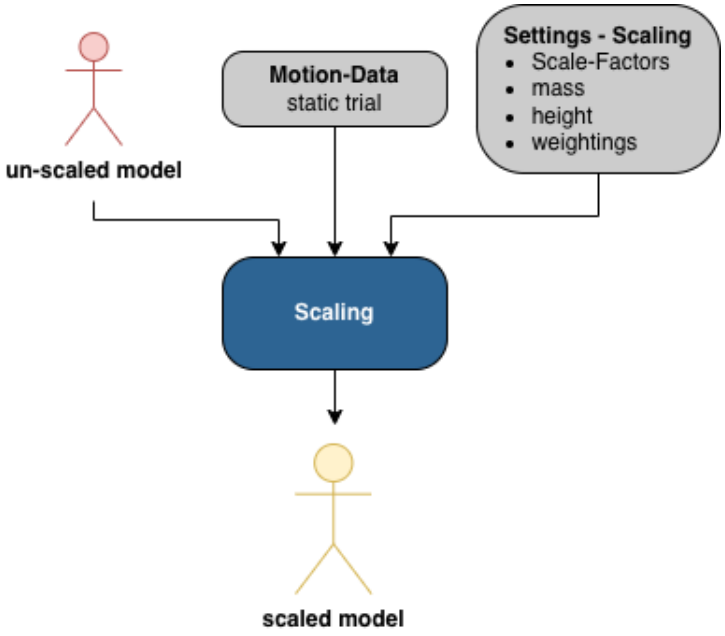


Fig. 15: Inputs and output of the scaling step in OpenSim, transforming an unscaled generic model using static trial motion data and scaling settings into a subject-specific scaled model

The settings of weighting of markers and coordinates for the process of Scaling are listed in Table 21 in the appendix of this thesis.

2.3.2.2 Inverse Kinematics (IK)

IK (Fig. 16) has been applied using the formerly scaled model, motion data of the dynamic trial and marker's weightings (Table 23). IK have been applied on each of the formerly extracted ground contact events, hence, every repetition of the FL. They have been optimised iteratively in a high-quality effort approach for optimal results of IK because they serve as fundamental input for further instances of the post-processing pipeline.

The results of IK would be considered as reliable and of sufficient accuracy if the highest RMS error of all markers exhibited a value of below 2 cm (Butler et al., 2017).

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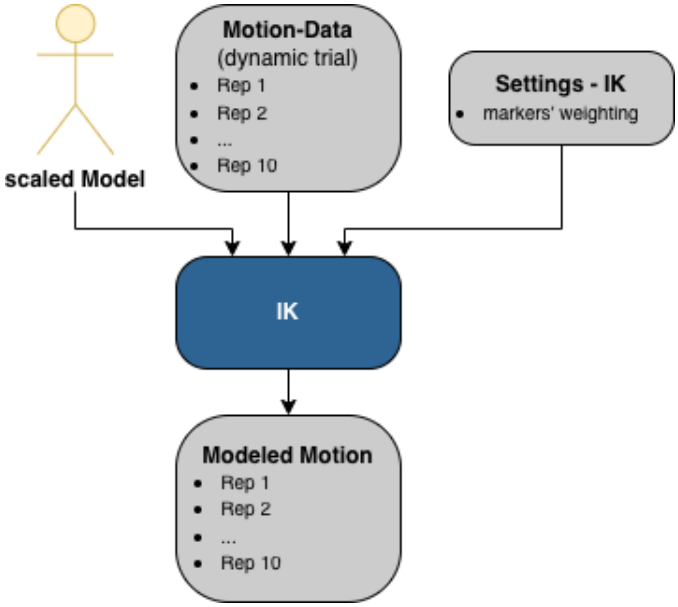


Fig. 16: Inputs and output of the IK processing step in OpenSim, including the scaled model, dynamic motion-capture data, and IK-settings, resulting in modelled kinematics

2.3.2.3 Inverse Dynamics (ID)

The setup of the built-in OpenSim tool of ID is made up of the formerly obtained IK solution, GRF data, the scaled model and appropriate settings of low-pass-filtering-frequency of coordinates (Fig. 17). The cut-off-frequency was set to 6 Hz, which is the recommendation of the official OpenSim Documentation (Hicks, 2018a).

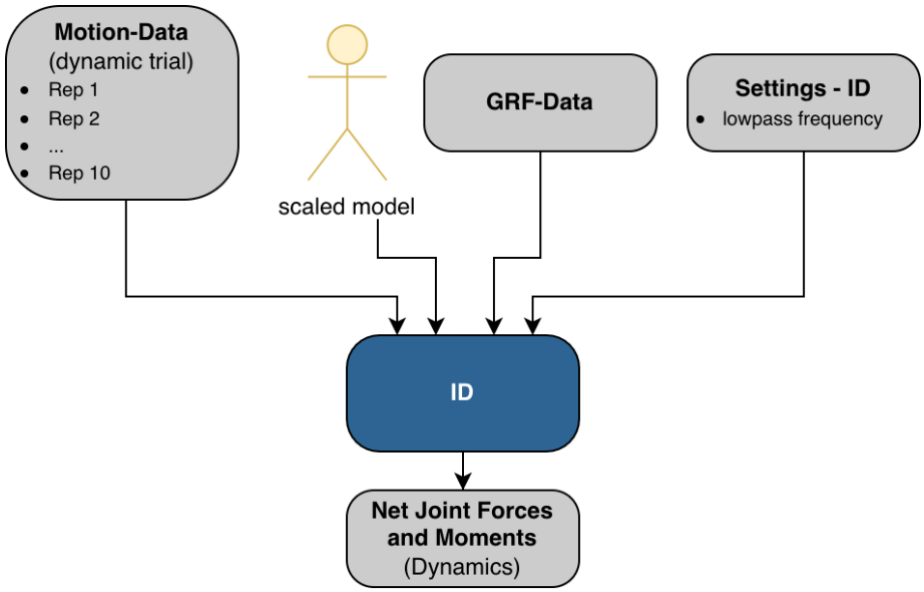


Fig. 17: Inputs and outputs of the ID processing step in OpenSim, including modelled kinematics from IK, a scaled model, GRF data, resulting in net joint moments for all DOFs

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2.3.2.4 Reduced Residual Algorithm (RRA)

In this study, RRA was applied as an iterative optimisation procedure following recommended best-practice guidelines provided in the OpenSim Documentation (Hicks, 2018c). Initially, the simulation was executed using default parameter settings. Subsequently, model parameters were systematically refined in a stepwise manner to minimise residual forces and moments while maintaining accurate kinematic tracking.

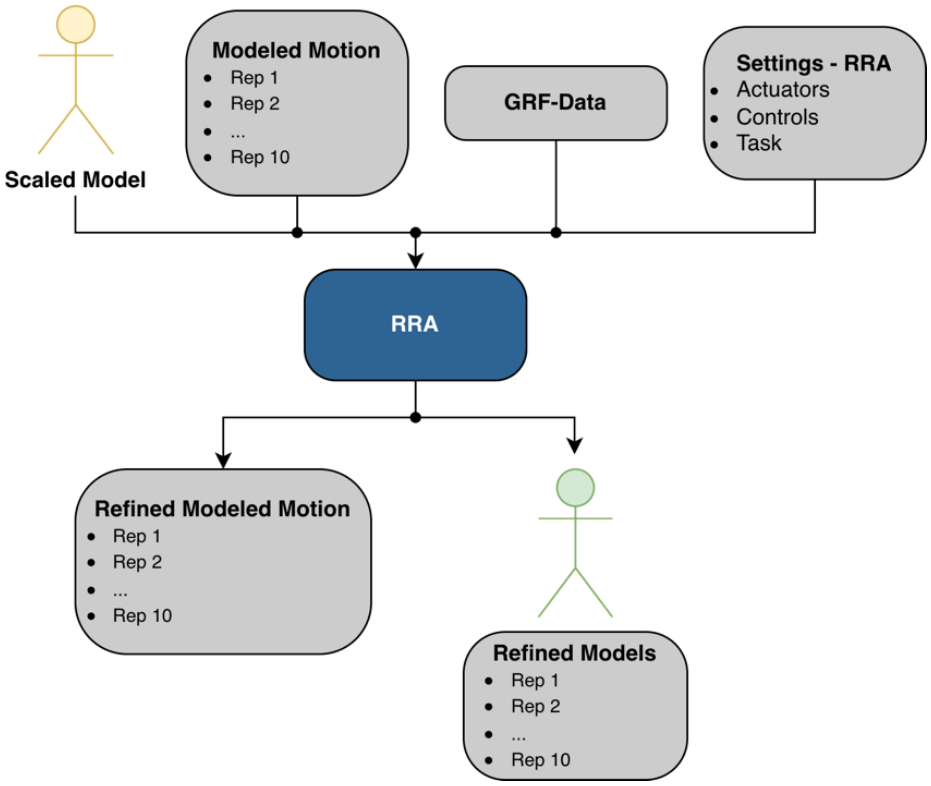


Fig. 18: Inputs and outputs of the RRA in OpenSim, including the scaled musculoskeletal model, motion from the IK solution, GRF data, RRA settings, resulting in refined kinematics and repetition-specific refined models

In a first attempt, tracking weights were adjusted such that coordinates with low tracking errors were assigned reduced weights, whereas coordinates exhibiting larger discrepancies were weighted more strongly to improve their agreement with experimental data. In a second attempt, the optimal strength of residual actuators was tuned in order to minimise their required action.

The optimisation process was considered converged when further parameter adjustments did not result in a meaningful reduction of residuals. Acceptance criteria were defined

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based on recommendation on established thresholds according to the OpenSim Documentation (Hicks, 2018c): maximum and RMS errors of rotational coordinates were required to remain below 2 °, while translational coordinate errors were constrained to be below 2 cm.

As illustrated in Fig. 18, the processing workflow resulted in a repetition-specific optimised motion and its corresponding adapted musculoskeletal model. The torso segment was selected for centre-of-mass coordinate adjustment, as it represents the body segment with the greatest mass and therefore exerts the largest influence on overall model dynamics. In addition, each generated model received individual mass-adjustment recommendations as part of the RRA output. These recommendations were subsequently implemented to adapt the mass properties of the respective models for further simulation procedures. This procedure was performed in accordance with the official OpenSim documentation guidelines.

2.3.2.5 Static Optimisation (SO)

As priorly with the RRA procedure, SO (Fig. 19) was applied as an iterative optimisation process in accordance with best-practice guidelines provided in the OpenSim documentation (Hicks, 2018d). The initial simulation was performed using the optimised residual actuator settings obtained from the RRA step. To complete the actuator set, reserve torque actuators were introduced for the hip, knee, and ankle joints of both limbs, initially assigned an optimal strength of 5 Nm.

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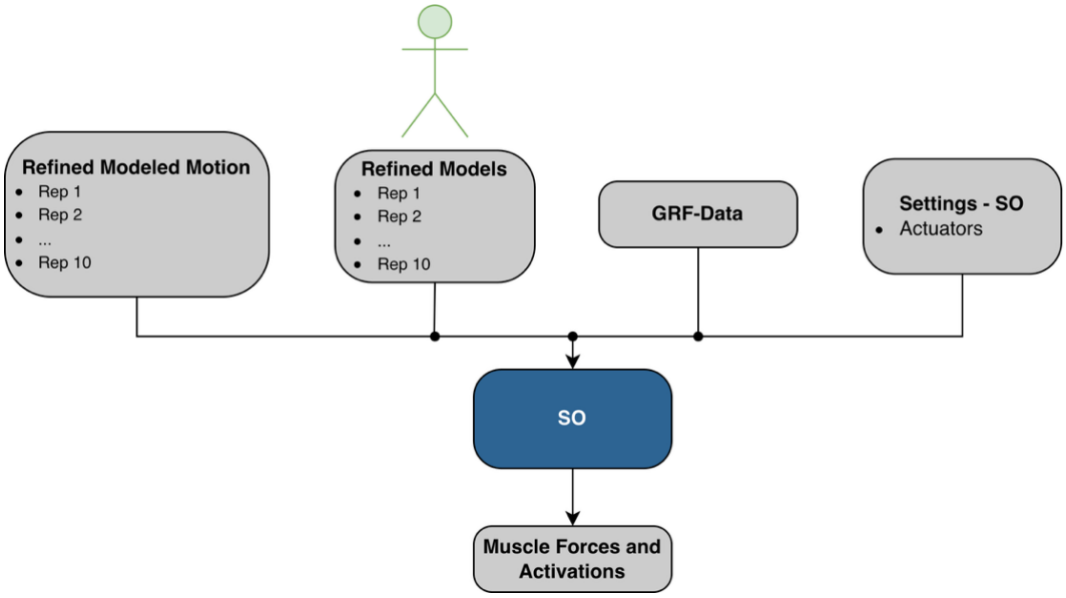


Fig. 19: SO tool in OpenSim using refined kinematics and models from RRA, GRF data, and SO settings to estimate muscle forces and activations

Subsequently, both model and actuator parameters were systematically refined to ensure a stable and physiologically plausible solution across all time frames. If muscle control signals reached their upper bound, indicating that muscles were operating at their maximum isometric force, the model's overall strength was increased by 10 %. If this adjustment remained insufficient to satisfy all constraints, the strength of the reserve actuators was adapted. In line with OpenSim recommendations, the contribution of reserve actuators was limited such that their peak torques did not exceed 10 % of the corresponding net joint moments derived from the inverse dynamics analysis (Hicks, 2018d).

Repetitions that continued to exhibit disproportionately high residual or reserve actuator contributions after this optimisation process were considered to contain artefacts and were therefore excluded from further analysis. A simulation was deemed valid when a stable solution was achieved, muscle activation values remained within physiological bounds (0–1) throughout the trial, and reserve actuator contributions remained within the defined thresholds.

2.3.2.6 Joint Reaction Analysis (JRA)

A JRA (Fig. 20) has been performed as a further consecutive step to calculate for the metric of interest, the KAM. To ensure consistency between processing steps, particular care has been given to align the JRA setup with the preceding SO analysis. This included using

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identical model configurations, actuator definitions — including residual and reserve actuators — muscle force outputs obtained from SO, and numerical integration settings such as step interval. Maintaining this consistency is essential for obtaining valid and interpretable joint reaction force estimates (Hicks, 2019a).

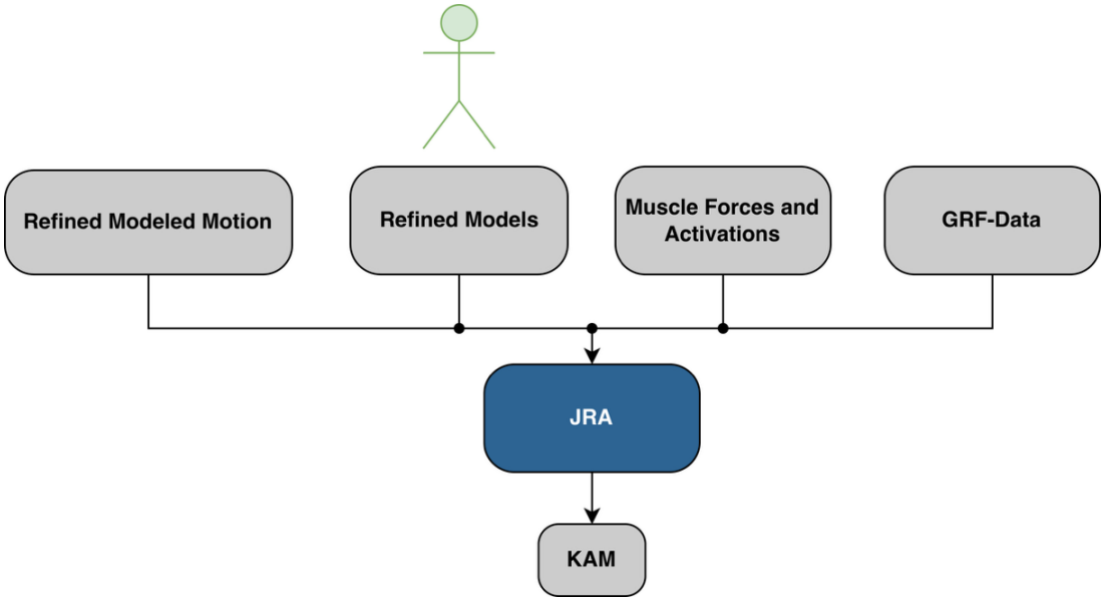


Fig. 20: Inputs and output of the JRA tool in OpenSim, including refined kinematics and models from RRA, GRF data, and muscle forces and activations from SO, resulting in the KAM

2.3.3 Surface Electromyography (sEMG)

sEMG of the GMED of the dominant side was analysed during both MVIC and forward lunge trials to investigate potential fatigue-related changes in muscle activation behaviour. Signal processing and analysis were performed separately for MVIC and dynamic trial recordings using custom-written Python scripts (Version 3.13.7) (Python Software Foundation, 2025).

2.3.3.1 Maximum Voluntary Isometric Contraction (MVIC)

For the analysis of the sEMG activity of the GMED of the dominant side during MVIC, a PRE-to-POST comparison was performed, with PRE representing the rested condition and POST representing the fatigued condition following the fatigue intervention. As described previously in Section 2.2.6 MVIC, two MVIC attempts were recorded for both conditions, separated by a resting period between attempts. All sEMG data were acquired at a

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sampling frequency of 1.2 kHz and processed using custom-written scripts in Python (Python Software Foundation, 2025).

Following acquisition, the raw sEMG signal underwent preprocessing consisting of demeaning, fourth-order Butterworth bandpass filtering (20–450 Hz), and 50 Hz notch filtering to reduce power-line interference. Subsequently, the signal was full-wave rectified and low-pass filtered (6 Hz, fourth-order Butterworth) to generate a smoothed linear envelope for burst detection. Burst detection was performed automatically on the rectified and smoothed signal. To improve robustness against artefacts and false-positive detections, a minimum burst duration of 1.0 s was defined. For each participant and condition separately, the 95th percentile of the processed signal was used as a reference value for threshold determination. Based on this reference, hysteresis thresholds were defined using a 70 % threshold for burst onset detection and a 25 % threshold for burst termination. PRE and POST recordings were processed independently while applying identical filtering procedures.

For frequency-domain analysis, the signal underwent demeaning, fourth-order Butterworth bandpass filtering (20–450 Hz), and 50 Hz notch filtering, without rectification or low-pass filtering. Power spectral density (PSD) analysis was subsequently performed for each detected MVIC burst using Welch's method with a Hamming window (segment length: 1024 samples, overlap: 512 samples, FFT length: 1024 samples). Following PSD estimation, both MVIC bursts of each condition were averaged. Subsequently, PSD spectra were averaged across all participants to generate a group mean PSD spectrum for the PRE and POST conditions, respectively. MNF was then extracted from the corresponding mean PSD spectrum and used as the primary fatigue-related outcome parameter. MNF was selected as an indicator of muscular fatigue.

2.3.3.2 Trial

For the analysis of the GMED sEMG during the trial of the FL, a PRE-to-POST comparison was conducted to investigate potential fatigue-related changes in muscle activation magnitude and activation behaviour. As mentioned in the previous section, PRE represented the rested condition, whereas POST represented the fatigued condition following the FE. All sEMG data were acquired at a sampling frequency of 1.2 kHz and processed using custom-written scripts in Python.

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Initially, the raw sEMG signal was extracted from the corresponding trial .c3d files. In addition, the PRE MVIC recording of the GMED was imported to establish a reference signal for subsequent MVIC normalisation (%MVIC). Segmentation of the trial sEMG signal was performed based on ground-contact events derived from vertical ground reaction force (vGRF) data, using the methodology described previously in Section 2.3.1 Immediate post-processing. Although the sEMG system and the motion capture/force-plate system sampled data asynchronously, both systems operated using the same acquisition frequency and clock source. Consequently, temporal misalignment between systems was assumed to originate primarily from minor data-transfer latency. As this latency was previously determined to be within the millisecond range and therefore negligible relative to the duration of the analysed movement phases, the datasets were considered sufficiently synchronous for the purposes of the present analysis.

To establish the MVIC reference signal, the PRE MVIC recording underwent preprocessing consisting of demeaning, fourth-order Butterworth bandpass filtering (20–450 Hz), 50 Hz notch filtering, full-wave rectification, and fourth-order Butterworth low-pass filtering at 6 Hz. Burst detection was subsequently performed using the same methodology described in Section 2.3.3.1 Maximum Voluntary Isometric Contraction (MVIC. Following burst identification, a moving average window of 500 ms was applied within the detected bursts to improve robustness against transient spikes and signal artefacts. The maximum value obtained across the moving-average windows of both PRE MVIC bursts was defined as the individual 100% MVIC reference value. This PRE-derived reference value was subsequently used for normalisation of both PRE and POST FL trials.

The PRE and POST trial sEMG signals underwent identical preprocessing and were subsequently segmented according to the detected ground-contact events. Thereafter, all segments were amplitude-normalised to the previously established 100% MVIC reference value and time-normalised to 101 data points representing 0–100% of the analysed movement phase. Following normalisation, all repetitions within a trial were averaged to generate subject-specific mean activation profiles for the PRE and POST conditions, respectively. Subsequently, group mean activation profiles were calculated across all participants for descriptive and comparative analysis.

For quality assurance and outlier detection, the mean PRE and POST activation profiles of all participants were visually inspected using combined plots. In addition, statistical outlier detection was performed based on the 1.5 x interquartile range criterion. Participants

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identified as outliers according to this criterion were excluded pairwise from both the PRE and POST conditions, as well as from the corresponding MVIC analysis. Furthermore, participants exhibiting missing activation patterns, or physiologically implausible activation profiles were excluded from both trial and MVIC sEMG postprocessing.

2.4 Validation

2.4.1 Fatigue

As the successful induction of targeted GMED fatigue represented a prerequisite for achieving the aims of the present thesis, a shift of the MNF towards lower frequencies was interpreted as an indicator of muscular fatigue. In addition, fatigue-related alterations were further assessed by comparing PRE and POST GMED activation patterns during the FL task.

2.4.2 Model

The modelling process is validated by means of comparison of final as well as intermediate results like kinematics, net-joint-moments and KAM of the FL to comparable ones found in scientific literature.

Moreover, mechanical consistency of joint reaction analysis is being examined. This is done by a first robust validation step by the exploitation of known properties of the model. The ground-pelvis-joint for instance is modelled as a free joint. To validate mechanical consistency those reaction forces are checked if they show a permanent non-existent behaviour. If they do means strong evidence that the force propagation through the kinetic chain is internally consistent. A second validation step uses joint mechanics. The hip joint for instance, which is modelled as a ball-and-socket-joint resists translation but allows rotation. Therefore, the reaction moments of the hip are checked to be zero. Verifying that the moment components at the hip are zero confirms that joint constraints and force calculation are behaving as expected. These checks imply confidence that the remaining joint reaction forces such as at the ankles and the knees are consistent with the muscle forces resulting from the static optimization analysis (Hicks, 2020).

2.4.3 Surface Electromyography (sEMG)

To validate the measured sEMG of the GMED of the dominant side, mean activation during the FL is being compared with available data from scientific literature.

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3 Results

A total of ten participants were included in the study and successfully completed the laboratory session. Based on the EHIS-PAQ, 70 % of participants were classified as sufficiently physically active.

Following data processing and quality control procedures, two datasets were excluded from specific analyses. One participant's motion-capture dataset was excluded from the biomechanical analyses due to non-convergence of the musculoskeletal model solver during the estimation of muscle activation patterns. In addition, the sEMG dataset of a different participant was excluded from the EMG analyses due to the presence of non-physiological signal artefacts. Exclusions were applied only to the affected analyses; all remaining valid data from these participants were retained and included in the respective other analyses.

The following results represent group-averages. The underlying participant-specific data is the average across repetitions or bursts for the FL exercise and the MVIC of the GMED respectively.

3.1 Knee Adduction Moment (KAM)

Nine out of ten datasets have been used for this analysis. One participant had to be excluded because of non-convergence of muscle-activation-patterns during SO.

The KAM profiles (Fig. 21) for the Pre- and Post-fatigue conditions exhibit a rather similar temporal pattern during the normalised ground-contact phase. Both curves show a steep initial increase within the first 10 % of normalised time, followed by a more gradual rise until approximately 65 % of the ground-contact window. Thereafter, the KAM decreases progressively towards foot-off.

During the initial 10 % of normalised time, Pre- and Post-fatigue curves are nearly identical. From this point onwards, both trajectories remain consistently offset until the occurrence of the peak values, observed at 0.2931 ± 0.0626 Nm/kg (Pre-fatigue) and 0.2869 ± 0.0596 Nm/kg (Post-fatigue) (Table 5). Following the peak, the two curves converge continuously, with differences diminishing towards the end of the ground-contact-phase.

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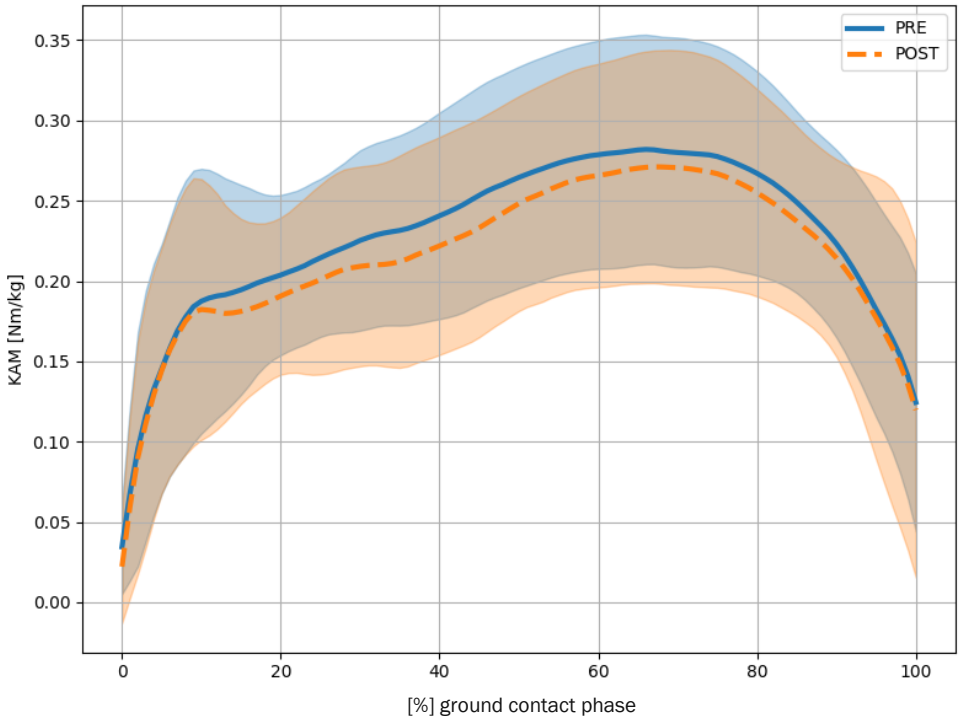


Fig. 21: Mean ± SD pre- and post-fatigue KAM during the FL for the functionally dominant leading leg, normalised by body mass and duration of ground contact phase

Table 5: Peak, average and accumulated KAM averages and SD for pre- and post-fatigue Conditions

Timepoint	Peak[a] [Nm/kg]	Mean[b] [Nm/kg]	AUC[c] [Nm/kg]
PRE	0.2931 ± 0.0626	0.2273 ± 0.0574	0.2288 ± 0.0576
POST	0.2869 ± 0.0596	0.2153 ± 0.0574	0.2167 ± 0.0576

[a] average of subject-specific peak-KAMs

[b] arithmetic mean of over-all KAM

[c] area-under-curve cumulated KAM of individual subject averages KAM

One objective of this thesis was to provide normative data for the KAM during FL in healthy adults, Fig. 22 presents both the group-averaged profile and the corresponding participant-specific curves.

Overall, similar temporal KAM patterns were observed across participants. Individual profiles typically exhibited two local maxima, with the second peak showing the greater magnitude in most cases. The first peak occurred within the initial third of the ground-contact phase, with inter-individual variability in both magnitude and timing. A second, more pronounced peak was consistently observed at approximately 65 % of the ground-contact phase.

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The group-averaged curve results in a smoothed single-peak profile, incorporating the aggregation of individual double-peak structures. This curve can be conceptually divided into three phases: an initial steep increase during the first roughly 10 % of normalised time, a more gradual intermediate phase up to approximately 65 %, and a final decline towards foot-off.

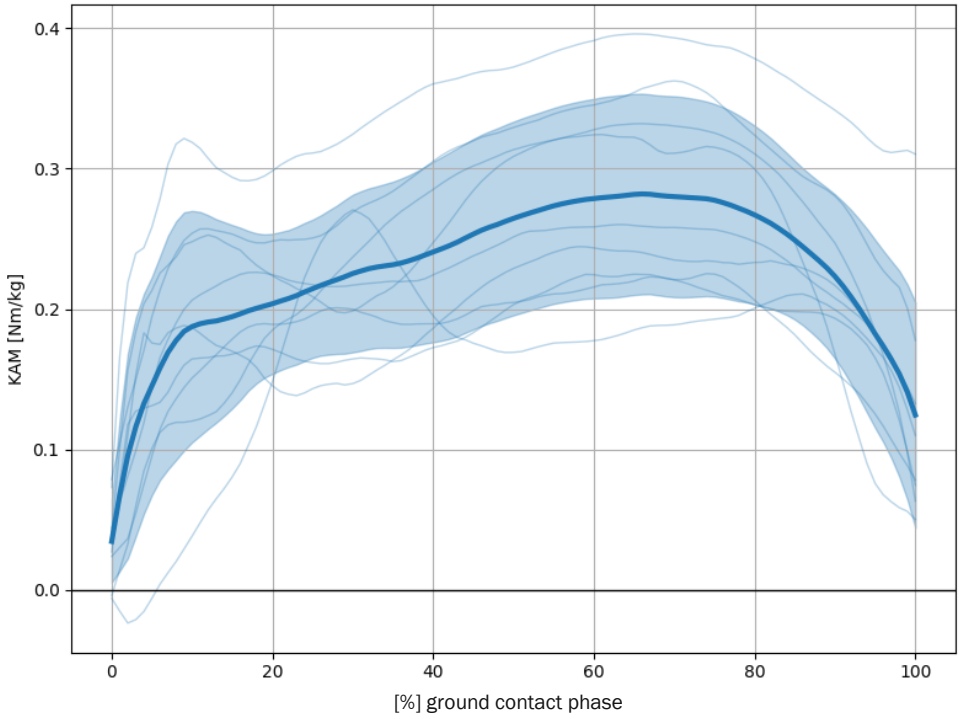


Fig. 22: Pre-fatigue KAM during the FL: group mean (bold line) and participant-specific trajectories (thin lines) with SD, normalised by body mass and duration of ground contact phase

3.2 Gluteus Medius – Surface Electromyography

Data of nine out of ten participants have been used for this analysis. One set of data had to be excluded because of the exhibition of non-physiological activation-patterns of the GMED.

The group-averaged sEMG activity of the GMED muscle during FL exhibited qualitatively similar temporal characteristics between the pre- and post-fatigue conditions. In both states, muscle activity increased during the initial 10 % of the normalised ground-contact phase. In the pre-fatigue condition, this increase transitioned into a plateau at approximately 15 %MVIC. In contrast, the post-fatigue condition exhibited a more

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pronounced activation pattern during the early ground-contact phase, suggesting a slightly more pronounced local maximum between approximately 10 % and 25 % of the normalised ground-contact phase, reaching about 22 %MVIC. This was followed by a plateau at a slightly elevated activation level of approximately 17 %MVIC, which additionally incorporated a second, less pronounced local maximum (Fig. 23).

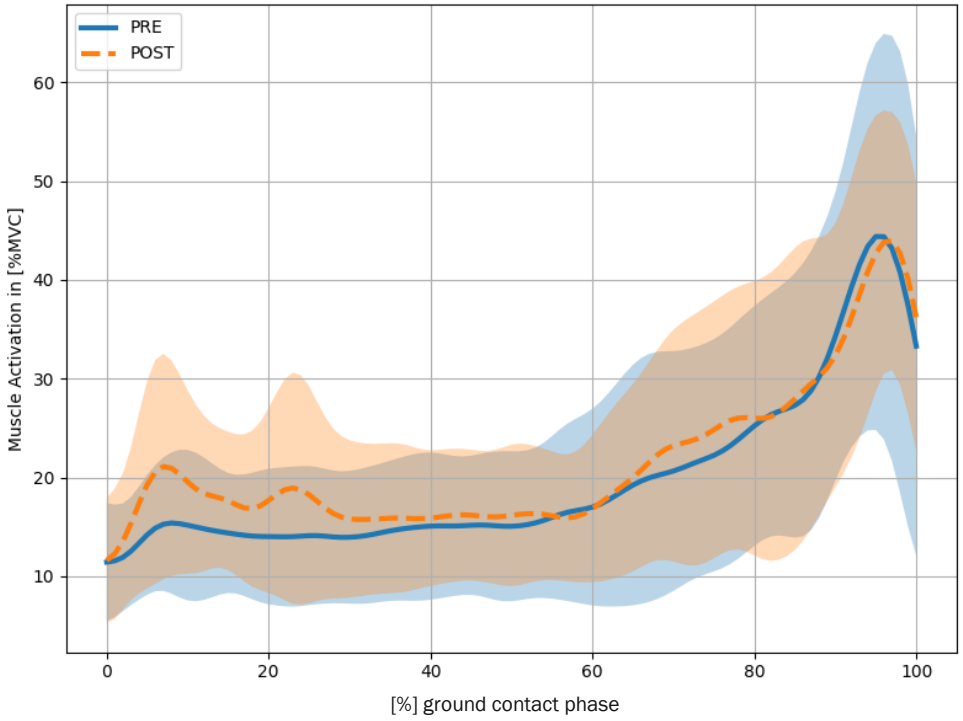


Fig. 23: Mean ± SD GMED activation during the FL: pre- and post-fatigue comparison across participants, normalised by duration of ground contact phase and by MVIC

Both steady-states continue to last till about 60 %MVIC of the ground-contact phase. From this time-point on, both curves exhibit similar activation levels with a moderate and nearly linear increase in activation in both conditions, reaching approximately 44 %MVIC of activity at around 95 %MVIC of normalised time. Around these maxima, both curves displayed an approximately symmetric local shape, followed by a decline in activation towards foot-off to values of roughly 35 %MVIC. Consistent with the high similarity of the group-averaged activation profiles, the mean peak activation derived from the individual participant curves differed only marginally between conditions, amounting to 48.75 ± 20.26 %MVIC in the pre-fatigue condition and 48.41 ± 13.54 %MVIC in the post-fatigue condition.

Peak activation showed a slightly higher activation in favour of the Pre-fatigue state, whereas both mean activation of group-averaged curves and accumulated activation across the ground-contact phase were slightly greater in the Post-fatigue condition. Mean

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activation amounted to 21.42 ± 8.57 %MVIC and 19.83 ± 8.30 %MVIC, while accumulated activation reached 21.39 ± 8.64 %MVIC and 19.80 ± 8.31 %MVIC, for the Post- and Pre-fatigue conditions, respectively (Table 6).

Table 6: Arithmetic mean, peak and accumulated GMED sEMG activation during ground contact phase of FL normalised to MVIC and time

Timepoint	Mean activation [a] [%MVIC]	Peak activation [b] [%MVIC]	AUC [c] [%MVIC*%time/100%]
PRE	19.83 ± 8.30	48.75 ± 20.26	19.80 ± 8.31
POST	21.42 ± 8.57	48.41 ± 13.54	21.39 ± 8.64

[a] arithmetic-mean of individual subject-means

[b] mean ± SD peak-activation of individual subject-peak-activations

[c] mean ± SD area-under-curve cumulated activation of individual subject-means

3.3 Gluteus Medius – Force

Nine of the ten available datasets were included in the analysis of GMED force during the ground-contact phase. Consistent with the KAM analysis presented in Section 3.1, one participant had to be excluded due to non-convergence of muscle activation patterns during the SO procedure.

The group-averaged GMED force profiles for the pre- and post-fatigue conditions exhibited a highly similar temporal pattern throughout the ground-contact phase (Fig. 24). Both conditions demonstrated a characteristic double-peak behaviour. Following a brief initial decline during the first approximately 3 % of normalised time, GMED force decreased from slightly above 1 N/kg to just below 1 N/kg, resulting in an early local minimum. Throughout this initial period, force values in the post-fatigue condition remained marginally greater than those observed in the pre-fatigue condition.

Subsequently, GMED force increased rapidly, reaching the first and more pronounced peak at approximately 15 % of the normalised ground-contact phase. Peak magnitudes were approximately 3.5 N/kg in both conditions, with greater values observed in the post-fatigue state. Following this initial peak, force declined progressively towards a local minimum occurring between approximately 50 % and 60 % of normalised time, where values reached roughly 1.5 N/kg.

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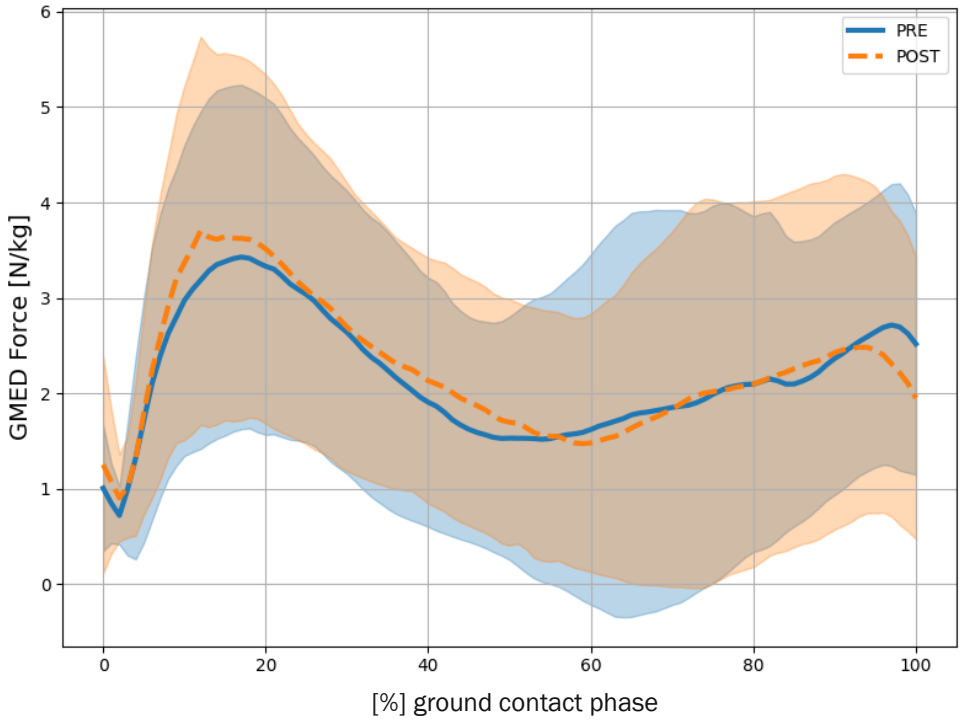


Fig. 24: Mean ± SD GMED force production during the FL for pre- and post-fatigue conditions, normalised by body mass and duration of ground contact phase

Thereafter, force increased again, resulting in a second local maximum at approximately 90–95 % of the ground-contact phase. The magnitude of this second peak was approximately 2.50 N/kg in the post-fatigue condition and 2.75 N/kg in the pre-fatigue condition. Following the second peak, GMED force decreased moderately towards foot-off, reaching values of approximately 2.0 N/kg and 2.5 N/kg for the post- and pre-fatigue conditions, respectively.

Collectively, results indicate marginally greater GMED force production in the post-fatigue condition with respect to both mean and peak force (Table 7). Mean force, calculated from the individual participant averages, amounted to 2.02 ± 1.32 N/kg in the pre-fatigue condition and 2.27 ± 1.42 N/kg in the post-fatigue condition. Similarly, peak force,

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calculated as the mean of the individual participant maxima, increased from 4.13 ± 2.02 N/kg pre-fatigue to 4.26 ± 2.07 N/kg post-fatigue.

Table 7: Arithmetic mean and peak GMED force during ground contact phase of FL exercise normalised to body mass and time

Timepoint	Mean force [a] [N/kg]	Peak force [b] [N/kg]
PRE	2.02 ± 1.32	4.13 ± 2.02
POST	2.27 ± 1.42	4.26 ± 2.07

[a] arithmetic-mean of individual subject-means

[b] mean ± SD peak-activation of individual subject-peak-forces

3.4 Gluteus Medius – Fatigue

Nine out of ten sets of data, which have previously been examined under 3.2 Gluteus Medius – , have been used within this processing step.

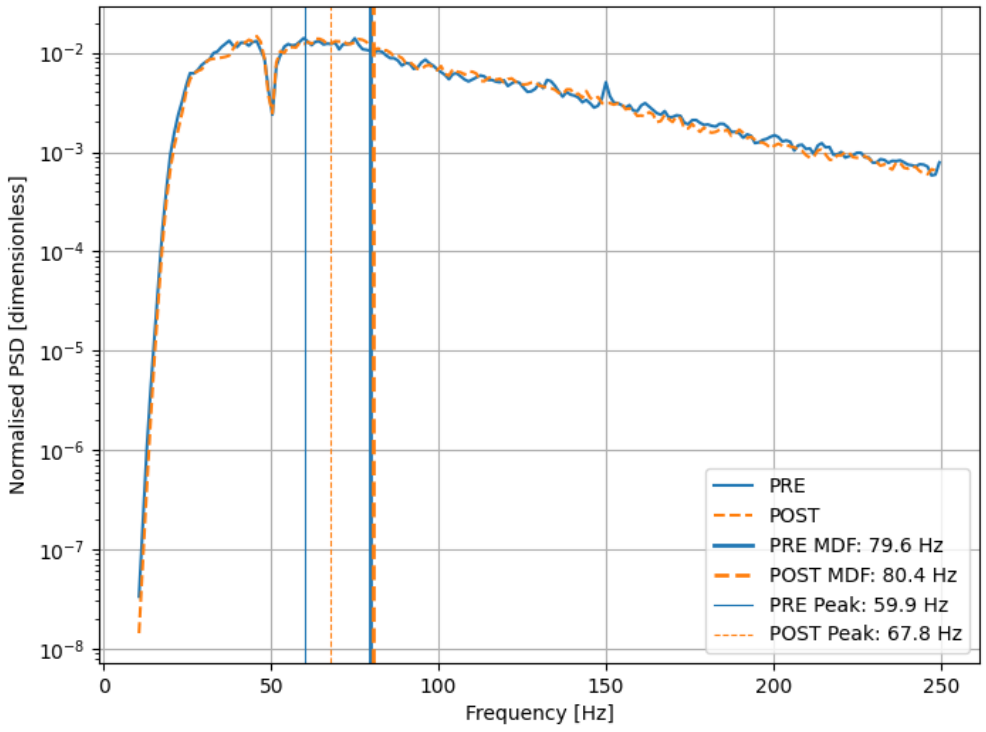


Fig. 25: Mean PSD profiles of functionally dominant GMED sEMG activity during MVIC with Median Frequency Markers

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Power Spectral Density (PSD) analysis of the GMED sEMG signal during MVIC revealed slight differences between the Pre- and Post-fatigue conditions with respect to spectral characteristics. The MNF increased from 78.58 Hz in the Pre-fatigue condition to 80.43 Hz in the post-fatigue condition. Similarly, the main frequency, defined as the frequency corresponding to the maximum PSD amplitude, shifted from 59.21 Hz in the Pre-fatigue state to 67.78 Hz following fatigue (Table 8).

Overall, the PSD profiles of both conditions exhibited highly similar spectral distributions and largely congruent curve morphology. In both spectra, a distinct attenuation around 50 Hz was observed, corresponding to the applied power-line interference filtering (Fig. 25).

Table 8: MVIC-MNF, peak values of PSD profiles and voltage RMS-values of pre- and post-fatigue conditions

Timepoint	MNF [Hz] [a]	Main Frequency [Hz] [b]	RMS [µV]
PRE	79.62	59.90	68.46
POST	80.43	67.78	73.37

[a] Median of arithmetic-mean-PSD across subjects

[b] Frequency of highest amplitude in PSD

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3.5 Kinematics

Data of all ten participants have been included within kinematics analysis.

Sagittal plane kinematics of the hip, knee, and ankle joints exhibited only minor differences between the Pre- and Post-fatigue conditions (Fig. 26). All three joints demonstrated a biphasic movement pattern consisting of an initial flexion phase during the eccentric portion of the movement, followed by an extension phase during the concentric portion. The transition between these phases occurred at approximately 60 % of the normalised ground-contact phase.

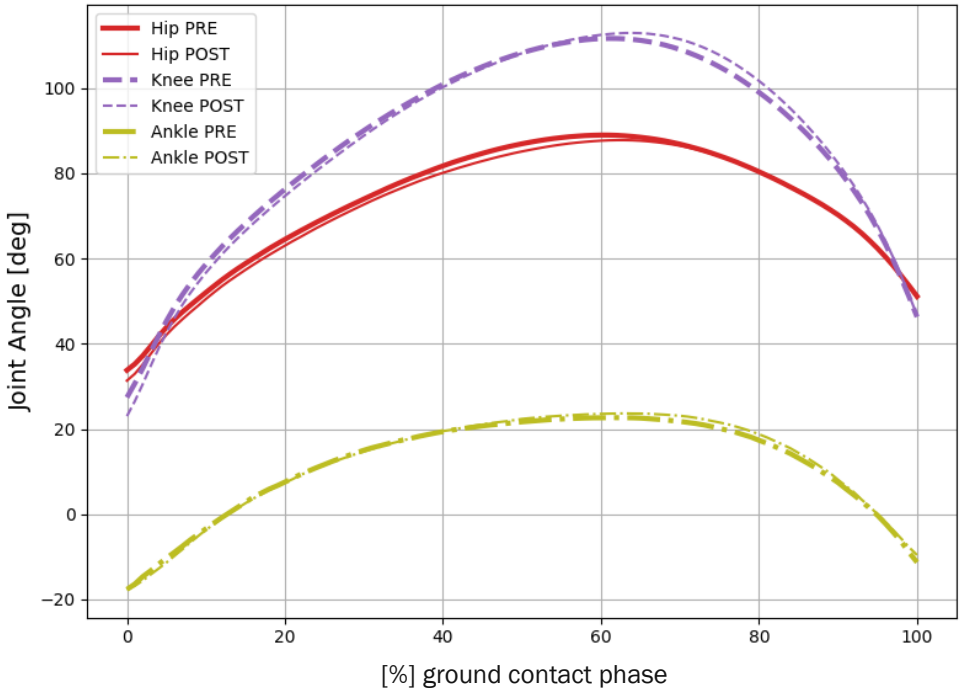


Fig. 26: Pre- and post-fatigue lower-limb kinematic profiles of the functionally dominant leading leg during the ground-contact phase

Peak joint angles were highly comparable between conditions (Table 9). Peak hip flexion angles amounted to 89.55 ± 7.30 ° and 88.27 ± 6.30 ° for the Pre- and Post-fatigue conditions, respectively. Corresponding peak knee joint angles reached 112.73 ± 7.85 ° and 113.86 ± 6.17 °, while peak ankle joint angles were 23.14 ± 9.06 ° and 24.02 ± 8.86 ° for the Pre- and Post-fatigue conditions, respectively.

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Sagittal plane ROM increased slightly in all three joints following fatigue. Hip joint ROM increased from 55.97 ± 5.44 ° in the pre-fatigue condition to 57.07 ± 6.81 ° post-fatigue. Knee joint ROM increased from 87.88 ± 5.86 ° to 92.58 ± 8.83 °, while ankle joint ROM increased marginally from 47.58 ± 7.66 ° to 47.94 ± 7.63 ° between the Pre- and Post-fatigue conditions.

Table 9: Peak joint angles and ROM of the hip, knee, and ankle during FL at pre- and post-fatigue-conditions

Joint	Timepoint	Peak [°]	ROM [°]
Hip	PRE	89.55 ± 7.30	55.97 ± 5.44
	POST	88.27 ± 6.30	57.07 ± 6.81
Knee	PRE	112.73 ± 7.85	87.88 ± 5.86
	POST	113.86 ± 6.17	92.58 ± 8.83
Ankle	PRE	23.14 ± 9.06	47.58 ± 7.66
	POST	24.02 ± 8.86	47.94 ± 7.63

3.6 Net-Joint-Moments

Data of all ten participants have been included within sagittal net-joint-moment analysis.

Sagittal plane net-joint-moments of the hip, knee, and ankle exhibited a broadly biphasic temporal pattern comparable to the observed kinematic profiles. All three joints demonstrated an initial steep increase in negative moment magnitude within the first 10 % of the normalised ground-contact phase, followed by a more gradual increase until approximately 70 % of normalised time. Thereafter, moment magnitudes progressively decreased towards the end of the ground-contact phase, followed by a rapid decline approaching foot-off (Fig. 27).

Overall, post-fatigue net joint moments generally exhibited lower magnitudes compared to the pre-fatigue condition across all joints. An exception was observed for the knee joint during the second phase of the movement, where the post-fatigue condition demonstrated slightly greater moment magnitudes relative to the pre-fatigue condition (Fig. 27b).

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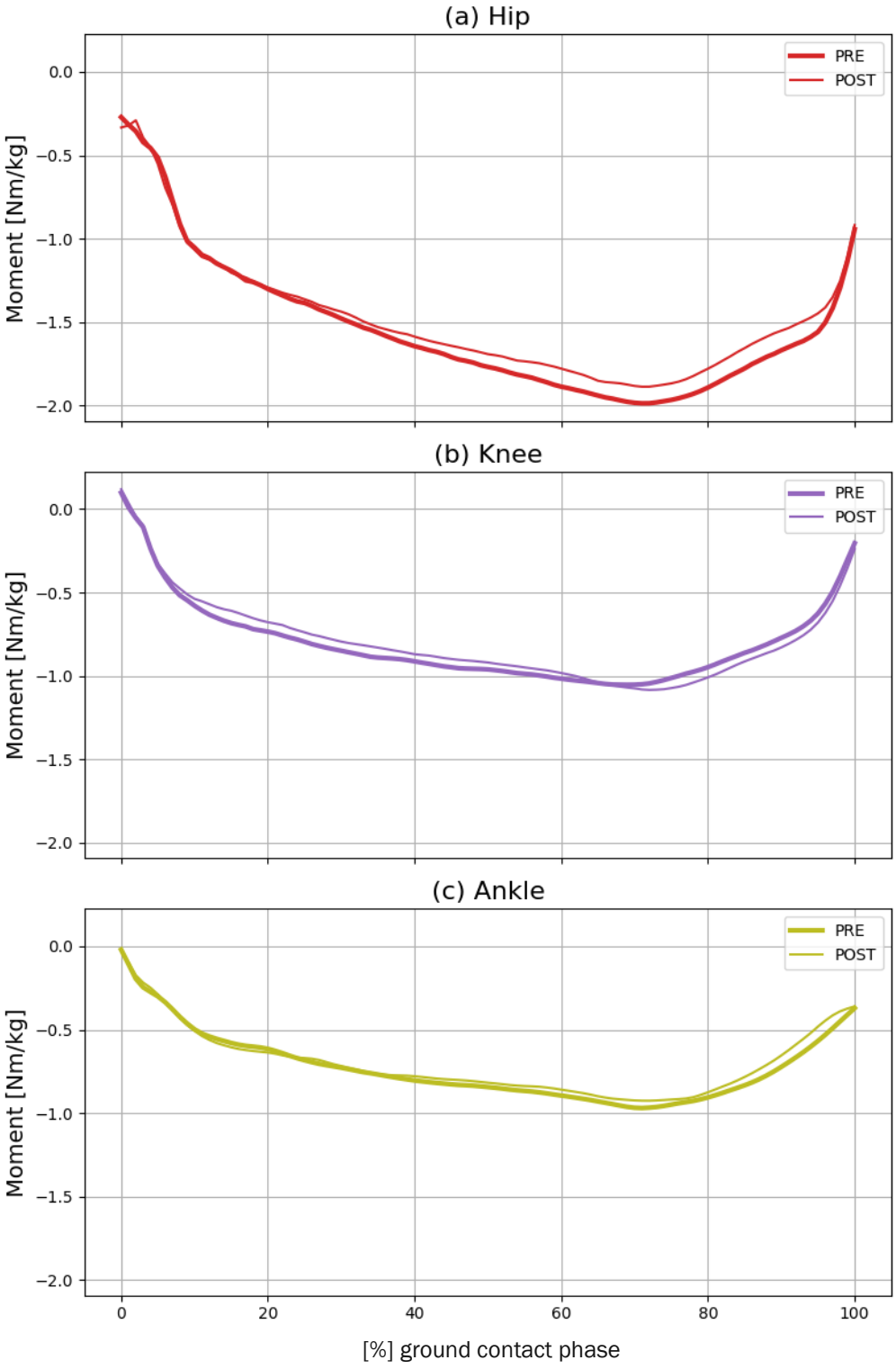


Fig. 27: Pre- and post-fatigue sagittal plane net joint moment profiles of the functionally dominant leading leg during the FL, normalised by body mass and duration of ground contact phase

Peak net joint moments at the hip amounted to -2.13 ± 0.14 Nm/kg in the pre-fatigue condition and -2.03 ± 0.16 Nm/kg post-fatigue. Corresponding peak knee-joint-moments

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were -1.23 ± 0.19 Nm/kg and -1.22 ± 0.12 Nm/kg for the Pre- and Post-fatigue conditions, respectively. Peak ankle-joint-moments reached -1.11 ± 0.16 Nm/kg in the pre-fatigue state and -1.06 ± 0.16 Nm/kg following fatigue (Table 10).

Mean net-joint-moments of the time-normalised curves were slightly reduced in the post-fatigue condition for all joints. Mean hip-joint-moments decreased from -1.53 ± 0.07 Nm/kg to -1.47 ± 0.11 Nm/kg, while mean knee-joint-moments decreased marginally from -0.80 ± 0.18 Nm/kg to -0.79 ± 0.13 Nm/kg between the Pre- and Post-fatigue conditions. Similarly, mean ankle-joint-moments declined from -0.72 ± 0.11 Nm/kg Pre-fatigue to -0.70 ± 0.11 Nm/kg post-fatigue, indicating a general reduction in sagittal plane net-joint-moment production following fatigue (Table 10).

Table 10: Pre- and post-fatigue sagittal plane peak and mean net joint moments of the functionally dominant leading leg during the FL normalised to ground-contact phase

Joint	Timepoint	Peak [Nm/kg]	Mean [Nm/kg]
Hip	PRE	-2.13 ± 0.14	-1.53 ± 0.07
	POST	-2.03 ± 0.16	-1.47 ± 0.11
Knee	PRE	-1.23 ± 0.19	-0.80 ± 0.18
	POST	-1.22 ± 0.12	-0.79 ± 0.13
Ankle	PRE	-1.11 ± 0.16	-0.72 ± 0.11
	POST	-1.06 ± 0.16	-0.70 ± 0.11

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4 Discussion

The present thesis aimed to address the previously mentioned gaps of current scientific literature (1.5 Thesis Goals) by providing normative KAM data of healthy male adults during the FL exercise and by investigating the influence of targeted GMED fatigue on KAM behaviour, thereby contributing to the evidence base for clinically informed rehabilitation planning and progression criteria.

As scientific literature currently lacks directly comparable data regarding the KAM during execution of the FL, further discussion on reliability and interpretation of the obtained KAM values has been performed via comparison with available data from other functional movement tasks. Table 11 presents Peak-KAM values reported for squatting, single-leg squatting, and gait, alongside the FL values obtained within the present study.

Han et al. (2013), who investigated the KAM during bilateral squatting with a neutral foot position and horizontal thigh alignment at the lowest squat position, reported Peak-KAM values highly comparable to those observed during the FL in the present thesis, with values of 0.293 ± 0.346 Nm/kg and 0.293 ± 0.062 Nm/kg, respectively. These findings may indicate comparable frontal-plane knee loading characteristics between both movement tasks under controlled execution conditions.

In contrast, Khuu & Lewis (2019) reported lower Peak-KAM values during single-leg squats, with values of 0.200 ± 0.25 Nm/kg for females and 0.190 ± 0.25 Nm/kg for males. These reduced values may suggest a more evenly distributed mediolateral knee joint loading pattern during the single-leg squat task compared to the FL observed in the present study.

Tang et al. (2022), who analysed the first Peak-KAM during gait, observed greater KAM values compared to those obtained during the FL (Table 11), with post-fatigue gait values also demonstrating a significant increase relative to the non-fatigued condition. These findings suggest that gait may impose greater frontal-plane knee loading than controlled rehabilitation and strengthening exercises such as the FL. Furthermore, the direction of fatigue-related effects differed between studies, as the present thesis demonstrated a rather unexpected reduction in Peak-KAM from pre- to post-fatigue assessment, whereas Tang et al. (2022) observed an expected increase following fatigue.

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Table 11: Comparison of peak KAM values from comparable movement tasks reported in the literature, including FL results from the present study

Source	Movement	Peak-KAM [Nm/kg]	
		Pre-Fatigue	Post-Fatigue
Han et al., (2013) [a]	Squat	0.293 ± 0.346	—
Khuu & Lewis (2019) [b]	Single Leg Squat	0.200 ± 0.25 (females) 0.190 ± 0.25 (males)	—
Tang et al., (2022) [c]	Gait	0.444 ± 0.182 (females) 0.340 ± 0.166 (males)	0.545 ± 0.157 (females) 0.501 ± 0.148 (males)
Current Thesis [d]	FL	0.293 ± 0.062	0.2869 ± 0.0596

[a] Horizontal squats (thigh horizontal in lowest position) with shoulder wide stand, healthy adults balanced in biological sex

[b] Performance till maximum controllable ROM (78.6 ± 8.2°), contralateral leg is kept aside performing leg

[c] At timepoint of first peak of vGRF; young, healthy participants balanced in biological sex

[d] Pre-fatigue data

Only two experimental studies investigating GMED sEMG during the FL exercise could be identified within the current literature. When considering mean GMED activation, Alkjær et al. (2020) reported lower values of 11.3 ± 8.3 %MVIC compared to the mean activation observed in the present study (19.83 ± 8.30 %MVIC), whereas Muyor et al., (2020) reported higher activation levels (30 ± 9 %MVIC). Table 12 is providing a comparative overview.

The lower activation values reported by Alkjær et al. (2020) may partially be explained by methodological differences in exercise execution. Although the FL task was generally comparable, neither step length nor movement rhythm were standardised within their protocol. In contrast, the present study prescribed a step length corresponding to individual leg length and imposed a fixed movement rhythm. As several participants subjectively perceived the prescribed step length as longer than preferred, this may have increased the muscular effort required to maintain movement control and execution consistency, thereby contributing to greater GMED activation amplitudes compared to Alkjær et al. (2020). Furthermore, Alkjær et al. (2020) did not clearly specify the temporal boundaries used for calculation of mean activation values. In the present study, mean activation was determined strictly within ground-contact windows defined by vGRF

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thresholds, thereby improving temporal standardisation and comparability across repetitions.

Muyor et al., (2020), in contrast, reported higher mean GMED activation values of 30 ± 9 %MVIC during the FL exercise. A likely explanation for these elevated activation levels is the additional external loading applied during execution, corresponding to 60 % of the participants' five-repetition maximum.

Similar findings were observed for peak GMED activation. Alkjær et al. (2020) reported lower peak activation values of 28.4 ± 17.9 %MVIC compared to the peak activation observed in the present thesis (48.75 ± 20.26 %MVIC). Again, differences in exercise standardisation and the absence of clearly defined temporal analysis windows may account for these discrepancies between studies.

Table 12: Comparison of mean and peak GMED EMG activation (sEMG) during the FL reported in the literature, including results from the present study, normalised to MVIC

Source	Mean [%MVIC]	Peak [%MVIC]
Alkjær et al., (2020) [a]	11.3 ± 8.3	28.4 ± 17.9
Muyor et al., (2020) [b]	30 ± 9 [c]	–
Current Thesis [d]	19.83	44.39

[a] FL executed with a self-selected pace, values refer to the healthy control group

[b] FL executed using additional weight (60% of 5 repetition maximum)

[c] Information extracted out of a diagram since no explicit numeric data was provided

[d] Pre-fatigue data

As the current scientific literature provides only limited data regarding GMED contraction forces during the FL, the obtained results were compared with biomechanically similar movement tasks reported in previous studies (Table 13).

One such task is stair ascent, which exhibits a qualitatively similar force profile characterised by two force peaks, with the first peak being more pronounced, as observed in the present FL data. Bennett et al., (2018) reported mean GMED contraction forces of 3.83 ± 1.28 N/kg and 5.99 ± 1.97 N/kg in healthy individuals exhibiting varus and valgus knee alignments, respectively. These values exceed the mean GMED force observed in the present study (2.20 ± 1.32 N/kg) by approximately a factor of two to three, suggesting that stair ascent generally imposes greater demands on the GMED than the FL. However, the Peak-GMED force observed during the FL (4.13 ± 2.02 N/kg) approaches the mean force

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values reported during stair ascent, indicating that the weight-acceptance phase of the FL may require comparable levels of GMED force production.

Even greater GMED forces have been reported during more demanding dynamic tasks. Ueno et al., (2020), for example, observed mean GMED contraction forces of 12.52 ± 3.06 N/kg during drop vertical jumps, exceeding the values obtained in the present study by approximately a factor of six. This substantial difference is likely attributable to the markedly higher mechanical demands and impact forces associated with landing tasks compared to slowly executed exercises such as the FL variation chosen within this thesis.

Taken together, these comparisons suggest a relationship between GMED contraction force and overall movement intensity, with more demanding tasks requiring greater hip abductor force production. Furthermore, the findings of Bennett et al., (2018) indicate that static frontal-plane knee alignment may influence GMED force generation, as individuals with valgus alignment exhibited considerably greater GMED forces than those with varus alignment. This factor was not explicitly assessed within the present cohort and may therefore represent a potential source of inter-individual variability.

Nevertheless, the absence of directly comparable FL studies, together with differences in task characteristics, and participant-specific factors across studies, limits the extent to which external validation of the present GMED force results can be achieved.

Despite the observed differences in quantitative behaviour between the measured GMED sEMG activity (Fig. 23) and the simulated GMED muscle force (Fig. 24), both signals exhibited comparable temporal characteristics throughout the ground-contact phase. In both cases, a pronounced increase was observed during approximately the first 15 % of the normalised ground-contact window, followed by a steady-state (sEMG) and a reduction (GMED force). Subsequently, both quantities exhibited a second increase beginning at approximately 60 % of the ground-contact phase, reaching a local maximum at around 90 %, before decreasing towards the end of stance. This qualitative agreement in the temporal evolution of the two signals suggests that the simulation reproduces the general timing of GMED function during the FL and therefore provides qualitative support for the physiological plausibility of the estimated muscle forces.

In contrast, quantitative differences in the amplitudes of the measured sEMG activity and simulated muscle force are not necessarily unexpected. First, the two quantities represent different physiological measures. Whereas sEMG reflects the measured electrical activity

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associated with muscle excitation, muscle force is the mechanical output of the muscle–tendon unit originating from the modelling approach. Consequently, muscle force cannot be expected to scale proportionally with the measured sEMG amplitude. Within the used musculoskeletal model *RajagopalLaiUhlrich2023* (Lai et al., 2017; Rajagopal et al., 2016; Uhlrich et al., 2022), muscles are represented using Hill-type muscle models described by Millard et al. (2013), in which muscle force is determined not only by muscle activation but also by the instantaneous mechanical state of the muscle–tendon unit. Specifically, active force generation is modulated by the force–length and force–velocity relationships of the muscle fibres, while tendon compliance, including nonlinear elastic behaviour and tendon slack, influences force transmission. In addition, changes in pennation angle and passive force contributions, which primarily depend on muscle fibre length, further affect the resulting muscle force (Rajagopal et al., 2016). Consequently, activation represents only one determinant of muscle force production, and differences in amplitude between measured GMED sEMG activity and simulated GMED muscle force therefore still seem to be physiologically plausible.

An additional explanation for the observed discrepancies between measured GMED sEMG activity and simulated muscle force might be due to inherent limitations of sEMG. Skin electrodes record the electrical activity only from a limited volume of the underlying muscle and may therefore not fully represent the activation of the entire GMED. Furthermore, contamination by electrical activity originating from adjacent muscles cannot be entirely excluded and may have influenced the recorded signal. In addition, the FL involved a substantial hip ROM (approximately 57°), with the electrodes being applied in a neutral hip joint position prior to data collection. Consequently, relative displacement between the skin-mounted electrodes and the underlying muscle tissue is likely to have occurred throughout the movement. Such tissue movement may alter the muscle volume sampled by the electrodes and thereby influence the recorded sEMG amplitude independently of changes in the actual muscle activation. Collectively, these measurement-related factors represent additional plausible sources of discrepancy between the measured sEMG activity and the simulated GMED muscle force.

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Table 13: Comparison of mean and peak GMED force across different movement tasks reported in the literature, including FL results from the present study

Source	Task	Mean [N/kg]	Peak [N/kg]
Bennett et al., (2018)	Stair Ascent	3.83 ± 1.28 [a] 5.99 ± 1.97 [b]	–
Ueno et al., (2020)	Drop Vertical Jump [c]	12.52 ± 3.06	–
Current Thesis [d]	FL	2.20 ± 1.32	4.13 ± 2.02

[a] Results of varus-cluster sub-group

[b] Results of valgus-cluster sub-group

[c] Participants performed a maximum vertical jump after dropping off from a height of 30 cm

[d] Pre-fatigue data

For validation and quantification of the fatigue intervention, the MNF of the sEMG signal was evaluated as the primary fatigue parameter, as MNF has been reported to represent a valid and reliable indicator of muscular fatigue (Mohseni Bandpei et al., 2014). In addition, sEMG voltage amplitudes were analysed to improve comparability with existing literature.

Contrary to the expected fatigue response, both MNF and post-to-pre sEMG amplitudes obtained during GMED MVIC increased following the fatigue protocol (Table 14). Therefore, the applied intervention did not induce the anticipated reduction in signal frequency and amplitude commonly associated with muscular fatigue. In comparison, Rabello et al., (2022) reported a post-to-pre sEMG amplitude ratio of 82.05 % during GMED MVIC, whereas the present study demonstrated an increase to 107.2 %. One potential explanation may relate to methodological differences between fatigue protocols. Rabello et al., (2022) utilised a side-lying exercise position, thereby increasing the contribution of gravitational loading throughout the movement. In contrast, the present study employed an upright standing position combined with elastic resistance bands. Due to the non-linear elastic properties of resistance bands, resistance increased predominantly near the end ROM. Observationally, participants markedly reduced movement velocity near the wall-contact boundary, suggesting that substantial resistance was present only during a limited portion of the movement cycle. Consequently, the protocol may have induced high subjective exertion without generating sufficient continuous muscular loading to produce measurable neuromuscular fatigue.

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A further explanation may involve familiarisation effects associated with MVIC testing. As MVIC represented an unfamiliar task for most participants, repeated exposure may have improved participants' confidence and ability to generate maximal force output during post-measurements. Frost et al., (2012) demonstrated greater within-day reliability in participants experienced in resisting specific external loads, supporting the possibility that the observed increase in MNF may partially reflect task familiarisation.

Comparable discrepancies were observed relative to the findings of Tang et al., (2022), who reported decreases in MNF of 13.4 % in females and 18 % in males following fatigue. In contrast, the present study demonstrated a slight increase in MNF of 1.02 %. This difference may again be attributable to substantial methodological differences between fatigue protocols. Tang et al., (2022) employed an endurance-oriented fatigue protocol consisting of repetitive side-lying hip abductions performed until failure without additional resistance, resulting in prolonged continuous loading durations averaging 685 s. In contrast, the present thesis utilised a comparatively high-resistance protocol with substantially shorter loading periods. Therefore, differences in contraction characteristics and local muscular metabolic demands may have influenced both the magnitude and nature of the induced fatigue response.

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Table 14: Comparison of GMED fatigue indicators across studies, including pre- and post-fatigue EMG during MVIC, abduction strength loss, and MNF changes

Source	Pre-MVIC-activation	Post-MVIC-activation	Change in MNF [%]
Rabello et al., (2022) [a]	1120.8 ± 530.3 mV	919.8 ± 592.1 mV	—
	Post-to-Pre-Ratio: 82,05 %		
Tang et al., (2022) [b]	—	—	- 13.4% (female) - 18% (male)
Current Thesis [c]	RMS: 68.46 µV	RMS: 73.37 µV	+ 1,02%
	Post-to-Pre-Ratio: 107,2 %		

[a] Highly similar fatigue protocol to this thesis' experiment, load: 10-repetition maximum, participant's position during performance: side-lying

[b] Endurance-based fatigue-protocol, load: without additional load, participant's position during performance: side-lying

[c] Load: on-the-fly adapted load (8-15-repetition maximum), participant's position during performance: upright standing position

The Peak-Joint-Angles obtained for the hip, knee, and ankle joints demonstrated high agreement with comparable values reported in previous literature (Table 15). The greatest similarity was observed relative to the findings of Farrokhi et al., (2008), likely due to the comparable execution constraints applied in both studies, particularly regarding step length standardisation.

Although Gao et al., (2023) investigated a modified FL execution, with a final position on the front force-plate after the execution, thereby limiting direct comparability of absolute ROM values, the general tendencies observed were consistent with the present findings. Specifically, both studies demonstrated increases in joint ROM for all three joints from pre- to post-assessment following the fatigue intervention.

More specifically, the present study demonstrated increases in peak ankle and knee joint angles accompanied by a simultaneous reduction in peak hip joint angles following the fatigue protocol. This movement pattern may indicate execution of the FL with a deeper lunge position while simultaneously maintaining a more dorsally tilted trunk posture with a posterior tilted pelvis. These adaptations may reflect a compensatory or potentially more mechanically economical movement strategy emerging in response to fatigue.

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Table 15: Observed peak hip, knee, and ankle joint angles during FLs across different studies, including results from the present study

Source	Timepoint	Peak Joint Angle [°]		
		Hip	Knee	Ankle
Farrokhi et al., (2008) [a]	–	87.4 ± 11.8	110.3 ± 5.9	25.3 ± 5.7
Gao et al., (2023) [b]	PRE	88.51 ± 12.36 (75.64 ± 5.74) [d]	−120.45 ± 10.45 (99.55 ± 7.71) [d]	32.05 ± 8.40 (32.53 ± 4.40) [d]
	POST	92.89 ± 9.23 (76.10 ± 9.53) [d]	−124.44 ± 8.34 (105.30 ± 8.27) [d]	24.56 ± 4.55 (32.78 ± 5.73) [d]
Current Thesis [c]	PRE	89.55 ± 7.30 (55.97 ± 5.44) [d]	112.73 ± 7.85 (87.88 ± 5.86) [d]	23.14 ± 9.06 (47.58 ± 7.66) [d]
	POST	88.27 ± 6.30 (57.07 ± 6.81) [d]	113.86 ± 6.17 (92.58 ± 8.83) [d]	24.02 ± 8.86 (47.94 ± 7.63) [d]

[a] Balanced in biological sex, normalised step-length (height of greater trochanter over ground), 6 seconds of duration per repetition (3 for each, concentric and eccentric)

[b] Males only, self-selected pace and step-length, end-position is on the front force-plate

[c] Males only

[d] ROM

The Peak-Net-Joint-Moments obtained for the hip, knee, and ankle joints demonstrated high agreement with values reported in comparable scientific literature (Table 16). The present data further indicated a consistent reduction in Peak-Net-Joint-Moments across all three joints from pre-to post-assessment. This observation may suggest a redistribution of vertical propulsion demands from the leading to the trailing limb following the fatigue intervention. Such an interpretation is supported by the previously discussed kinematic findings, which indicated a more posteriorly inclined trunk posture during post-trials: A posterior displacement of the total centre of mass (COM) may consequently increase the extensor moment demands of the trailing limb while reducing the net joint moments of the leading limb.

The observed pre-to-post reductions are partially consistent with the findings of Gao et al., (2023), who likewise reported reductions in knee and ankle net joint moments following

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fatigue. However, in contrast to the present study, Gao et al., (2023) observed an increase in Peak-Hip-Net-Joint-Moment. This discrepancy may be attributable to differences in FL execution between studies.

Furthermore, the results reported by Comfort et al., (2015) demonstrated generally comparable Net-Joint-Moment magnitudes, thereby further supporting the plausibility of the values obtained in the present thesis. Although Comfort et al., (2015) did not distinguish explicitly between dominant and non-dominant limbs, their findings nevertheless suggested the presence of interlimb differences (Table 16 [a]).

Table 16: Observed peak hip, knee, and ankle net joint moments during FLs across different studies, including results from the present study

Source	Timepoint	Peak-Net-Joint-Moment [Nm/kg]		
		Hip	Knee	Ankle
(Comfort et al., 2015) [a]		L: - 2.06 ± 0.32 R: - 2.12 ± 0.35	L: - 1.40 ± 0.13 R: - 1.30 ± 0.22	L: - 0.99 ± 0.16 R: - 1.02 ± 0.20
Gao et al., (2023) [b]	PRE	- 1.95 ± 0.53	- 1.54 ± 0.68	- 1.12 ± 0.13
	POST	- 2.34 ± 0.17	- 1.30 ± 0.30	- 1.06 ± 0.14
Current Thesis [c]	PRE	-2.13 ± 0.14	-1.23 ± 0.19	-1.11 ± 0.16
	POST	-2.03 ± 0.16	-1.22 ± 0.12	-1.06 ± 0.16

[a] L/R referring to the side of the execution (leading leg), Males only, no constraint on step-length or rhythm

[b] Males only, self-selected pace and step-length, end-position is on the front force-plate

[c] Males only

The present thesis aimed to provide normative biomechanical data regarding the KAM during execution of the FL exercise in healthy young physically active male adults and to investigate the influence of targeted GMED fatigue on lower-extremity biomechanics. A cohort of ten physically active male participants was included in the study. Due to technical and methodological limitations, one dataset had to be excluded from the EMG analysis because of signal artefacts, while another dataset was excluded from the KAM evaluation because of a non-converging musculoskeletal solver solution. Both datasets, however, remained suitable for the respective remaining analyses. Overall, the obtained biomechanical data demonstrated high agreement with comparable findings reported within current scientific literature, supporting the plausibility of the present results.

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Furthermore, all validation procedures described within the section 2.4 Validation were completed successfully, indicating satisfactory data quality and model performance throughout the workflow.

The present study provides novel normative KAM data for the FL exercise and thereby contributes to the currently limited evidence regarding frontal-plane knee loading during rehabilitation-related movement tasks. In addition, the obtained GMED activation amplitudes demonstrated that the FL exercise elicits substantial hip abductor activation, further supporting the recognised importance of the GMED as a frontal-plane stabiliser of the knee joint. Consequently, the present findings provide clinically relevant biomechanical information which may contribute to future rehabilitation guidelines and evidence-based clinical decision-making.

Regarding the fatigue intervention, the expected reductions in MNF, sEMG amplitude and GMED contraction force following the fatigue protocol could not be confirmed. Instead, slight increases in all three parameters were observed. These findings contrast with the results reported by Rabello et al., (2022), who applied a similar exhaustion protocol in a side-lying position, with methodological details regarding resistance application remaining insufficiently described. Differences in exercise position and resistance characteristics were therefore identified as potential contributors to the discrepant fatigue responses between studies. In addition, subjective participant feedback indicated that fatigue was perceived predominantly within the GMED of the standing, *non*-dominant limb rather than the dominant side targeted by the experimental design. Several participants further described the prescribed FL rhythm and step length as unintuitive or unnatural, potentially influencing movement execution and contributing to increased variability within the obtained data.

Kinematic analyses revealed increases in ankle and knee joint angles accompanied by reductions in hip joint angles following the fatigue protocol, suggesting the adoption of a deeper lunge position combined with a more posteriorly tilted pelvis, potentially representing a compensatory fatigue-related movement strategy. Correspondingly, kinetic analyses demonstrated reductions in Peak-Net-Joint-Moments across the hip, knee, and ankle joints during post-assessment, indicating potential alterations in propulsion strategies and posterior displacement of the centre of mass following fatigue. Furthermore, comparisons with existing literature suggested potential side-to-side differences in at least quantitative execution characteristics of the FL. Future studies

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should therefore aim to establish valid and reliable assessments capable of classifying lower limbs according to functional dominance in order to improve inter-study comparability and reduce methodological bias.

A particularly important aspect for future research concerns the inclusion of female populations within biomechanical investigations of rehabilitation. Females represent a substantial proportion of individuals sustaining ACL injuries and are consequently at elevated risk of developing PTOA. Moreover, women have been reported to utilise pain medication more frequently, exhibit poorer outcomes following ACLR and HTO, and are commonly present for total knee arthroplasty at older age, with greater OA severity levels, pain, muscular weakness, and functional limitations compared to men (Segal et al., 2024). Consequently, future studies should specifically investigate KAM behaviour, GMED activation characteristics, as well as fatigue responses, and movement adaptations during the FL exercise specifically in female populations in order to facilitate the improvement of the inter-sexual gap in patient-treatment-quality.

One major limitation of the present study relates to the standardisation constraints imposed on the execution of the FL. Approximately half of the participants reported that the prescribed stride length felt intuitively too long, while the externally imposed movement rhythm of four seconds per repetition was additionally perceived as unnatural like long. Consequently, it is possible that several participants did not perform the FL according to their individually preferred movement strategy. This may have reduced the participants' perceived level of movement control, which was also reflected in subjective participant feedback. As a result, the imposed movement constraints may have introduced additional movement variability and unintended fluctuations in the execution of the task, thereby contributing to increased variability and dispersion within the obtained biomechanical data.

A further limitation of the present thesis is the unequal representation of biological sex, as only male participants were included in the study cohort. This limitation is of particular relevance, given that females represent a substantial proportion of young individuals sustaining ACL injuries and are therefore at elevated risk of developing early PTOA (Larwa et al., 2021). Still, maintaining a highly homogeneous study cohort was considered a priority in order to minimise inter-subject variability and enhance the reliability and interpretability of the obtained results which resulted in a male-only cohort.

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The relatively small sample size of ten participants represents a further limitation of the present study, particularly given the observed inter-individual variability in the primary outcome measure. The selected sample size reflected a carefully considered compromise between methodological feasibility, including the substantial time requirements associated with data collection and processing, and the intended explanatory value of the obtained results. In this context, the methodological approach of previous comparable experimental studies was considered during study planning (Gao et al., 2023). Consequently, a cohort of ten participants was regarded as feasible while still providing sufficient data to establish preliminary normative values for healthy individuals and to identify potential biomechanical trends.

Another potential limitation of the present study relates to the applied fatigue protocol and its methodological constraints. Participant feedback consistently indicated that the fatigue exercise was highly demanding, therefore perceived subjectively as sufficient, with some participants terminating sets prematurely upon reaching a Borg scale rating of 19. In addition, most participants reported that fatigue was perceived predominantly within the GMED of the standing, and therefore non-dominant, limb. This response had not been anticipated during the planning phase of the study and may indicate that fatigue was induced on both sides with a possible higher effect on the non-dominant GMED rather than the solely intended dominant side.

A further potential limitation relates to the execution of the MVICs, as this contraction modality was unfamiliar to nearly all of the participants. Consequently, several subjects reported that the MVIC procedure initially felt unusual or unintuitive. This issue had been anticipated during the planning phase of the study and was therefore addressed by implementing a non-recorded familiarisation trial performed at approximately 50 % of the estimated MVIC. The purpose of this preliminary trial was to familiarise participants with the contraction form, countdown procedure, and general testing setup prior to the recorded measurements. Nevertheless, no formal assessment was conducted regarding the subjective effectiveness of this familiarisation procedure for the subsequent recorded MVIC trials.

The utilised musculoskeletal model *RajagopalLaiUhlrich2023* (Lai et al., 2017; Rajagopal et al., 2016; Uhlrich et al., 2022) may represent an additional limitation regarding movements involving large hip and knee joint ROM. Lai et al., (2017) reported that the model performs reliably for movements within approximately 120 ° of hip flexion and 140 °

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of knee flexion. Although these ranges were not exceeded within the present study, concerns emerged following one participant for whom no valid simulation solution could be obtained across multiple repetitions despite several physiologically justifiable optimisation attempts. This observation raised the possibility that the comparatively large ROM demands of the FL, relative to gait tasks for which the model was originally developed, may have pushed the model to its limits. However, as this issue occurred only in a single case, no definitive conclusions regarding the underlying cause can be drawn.

Nevertheless, substantial effort was dedicated to optimising the modelling workflow. This included the use of advanced scaling procedures (Di Pietro et al., 2024), including the projection of foot markers onto the horizontal ground plane to minimise scaling distortions, and the implementation of repetition-specific model adaptations during the RRA process. Furthermore, modelling of the trunk and upper extremities was included to reduce residual forces and provide a more comprehensive anthropometric representation.

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5 Conclusion

In summary, the present thesis expands the currently limited biomechanical evidence regarding the FL by providing normative data on KAM, GMED activation and contraction force, as well as kinematic and kinetic parameters in healthy young male adults. Although the intended fatigue effects could not be conclusively demonstrated, the findings contribute valuable information for future investigations examining targeted GMED fatigue and may support the development of evidence-based rehabilitation assessment and progression criteria.

An expected increase of dominant-side KAM was not observed. In contrast, the data suggest a reduction in KAM from pre- to post-assessment. This finding is in line with the observed increases in GMED activation and contraction force, which may indicate increased GMED engagement rather than a fatigue-induced reduction in neuromuscular capacity under the applied protocol.

Importantly, future research should specifically address female populations, as women are disproportionately affected by knee-related pathologies and often present with more severe clinical symptoms and outcomes.

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6 Appendix

Table 17: Complete list of utilised sEMG sensors, corresponding muscles, and body sides

Sensor#	Muscle	side
1	tensor fasciae latae muscle	dominant
2	long head of the biceps femoris muscle	dominant
3	long head of the biceps femoris muscle	non-dominant
4	semitendinosus muscle	dominant
5	medial head of the gastrocnemius muscle	dominant
6	lateral head of the gastrocnemius muscle	dominant
7	rectus femoris muscle	dominant
8	rectus femoris muscle	non-dominant
9	vastus lateralis muscle	dominant
10	vastus lateralis muscle	non-dominant
11	vastus medialis muscle	dominant
12	vastus medialis muscle	non-dominant
13	gluteus maximus muscle	dominant
14	gluteus maximus muscle	non-dominant
15	gluteus medius muscle	dominant
16	gluteus medius muscle	non-dominant

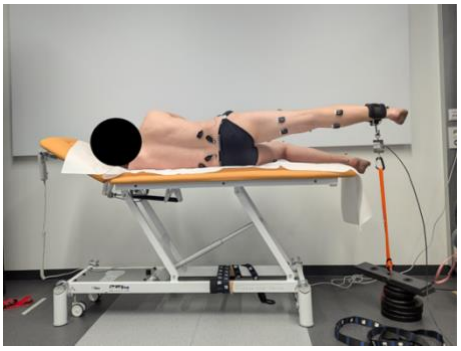
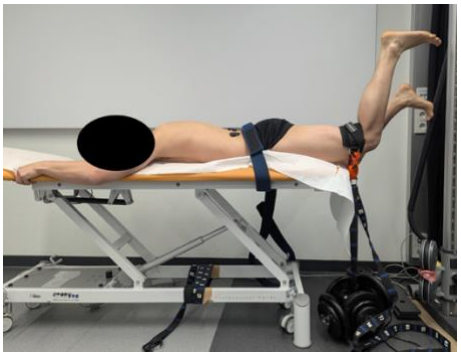
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Table 18: Order of MVIC tests, including assessment time point, muscle and body side

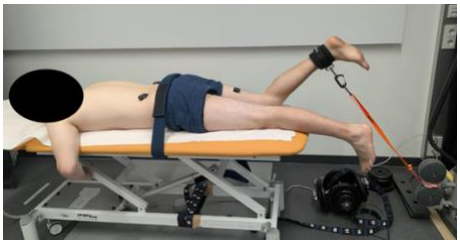
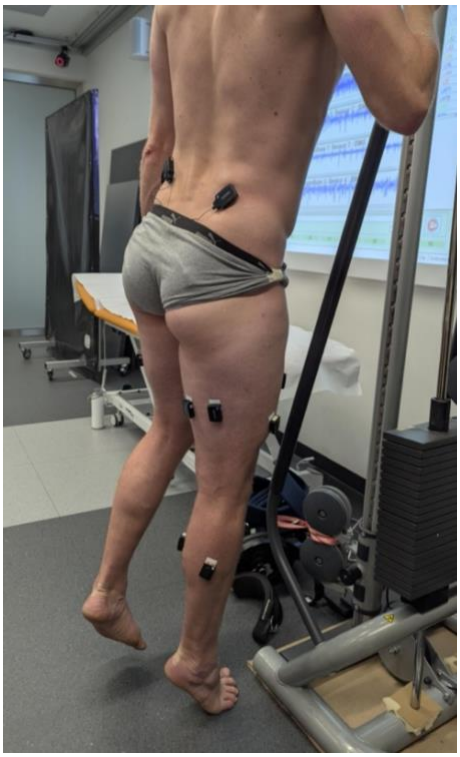
Testing-order	Pre/Post - Fatigue	Muscle	Side
1	pre	tensor fasciae latae muscle gluteus medius muscle	dominant
2	post	tensor fasciae latae muscle gluteus medius muscle	dominant
3	post	rectus femoris muscle vastus medialis muscle vastus lateralis muscle	dominant
4	post	medial head of the gastrocnemius muscle lateral head of the gastrocnemius muscle	dominant
5	post	rectus femoris muscle vastus medialis muscle vastus lateralis muscle	non-dominant
6	post	rectus femoris muscle vastus medialis muscle vastus lateralis muscle	dominant
7	post	long head of the biceps femoris muscle	non-dominant
8	post	long head of the biceps femoris muscle semitendinosus muscle	dominant
9	post	gluteus maximus muscle	non-dominant
10	post	gluteus maximus muscle	dominant

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Table 19: Documentation of testing positions and corresponding joint angles

Muscle	Testing Position	Joint Angles
tensor fasciae latae muscle gluteus medius muscle		side-lying 10° of hip-abduction 0° of hip-rotation 0° of knee-flexion
rectus femoris muscle vastus medialis muscle vastus lateralis muscle		upright sitting 45° of knee-flexion
gluteus maximus muscle		prone 10° of hip-extension 90° of knee-flexion pelvic fixation

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long head of the biceps femoris muscle semitendinosus muscle		prone 55° of knee-flexion pelvic fixation
medial head of the gastrocnemius muscle lateral head of the gastrocnemius muscle		upright unipedal stand maximum plantar-extension

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Table 20: Complete marker set including marker name (abbreviation), type, and corresponding anatomical landmark

Name	Explicit Name	Type	Anatomical Landmark
LACJ	left acromeo clavicular joint	anatomical	Most lateral Aspect of the left Clavicula
LASIS	left anterior superior iliac spine	anatomical	Prominence of left Anterior Superior Iliac Spine
LCC	left calcaneus	anatomical	Location of tendinous insertion of the left Achilles Tendon
LCL	left lateral condyle	anatomical	Left Lateral Femoral Epicondyle
LCM	left medial condyle	anatomical	Left Medial Femoral Epicondyle
LEL	Left elbow lateral	anatomical	Left Lateral Humeral Epicondyle
LEM	Left elbow medial	anatomical	Left Medial Humeral Epicondyle
LLAM	Left lower arm marker	anatomical	Left Dorsal Aspect of the Midshaft of the Ulna
LML	left lateral malleolus	anatomical	Left Lateral Malleolus
LMM	left medial malleolus	anatomical	Left Medial Malleolus
LMT1	left metatarso phalangeal joint 1	anatomical	Most Distal Aspect of the 1st Left Metatarsal Bone
LMT5	left metatarso phalangeal joint 5	anatomical	Most Distal Aspect of the 5th Left Metatarsal Bone
LPD1	left distal phalanx 1	anatomical	Centre of the 1st Left Toenail
LPSIS	left posterior superior iliac spine	anatomical	Prominence of left Posterior Superior Iliac Spine
LS1	left shank 1	tracking	Bottom Edge of the Cluster is placed at the lateral Aspect of the Left Shank one hand Proximal of the Left Lateral Malleouls Marker
LS2	left shank 2		
LS3	left shank 3		
LS4	left shank 4		
LT1	left thigh 1	tracking	Bottom Edge of the Cluster is placed at the lateral Aspect of the Left
LT2	left thigh 2		

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LT3	left thigh 3		Thigh one hand Proximal of the Left Lateral Epicondyle Marker
LT4	left thigh 4		
LTM	left trochanter major	anatomical	Prominence of the Left Major Trochanter
LTT	left tibial tuberosity	anatomical	Prominence of the Left Tibial Tuberosity
LWR	Left wrist radial	anatomical	Radial and most Distal Aspect of the Left Radius
LWU	Left wrist ulnar	anatomical	Dorsal and most Distal Aspect of the Left Ulna
RACJ	right acromeo clavicular joint	anatomical	Most lateral Aspect of the right Clavicula
RASIS	right anterior superior iliac spine	anatomical	Prominence of right Anterior Superior Iliac Spine
RCC	right calcaneus	anatomical	Location of tendinous insertion of the right Achilles Tendon
RCL	right lateral condyle	anatomical	Right Lateral Femoral Epicondyle
RCM	right medial condyle	anatomical	Right Medial Femoral Epicondyle
REL	Right elbow lateral	anatomical	Right Lateral Humeral Epicondyle
REM	Right elbow medial	anatomical	Right Medial Humeral Epicondyle
RLAM	Right lower arm marker	anatomical	Right Volar Aspect of the Midshaft of the Ulna
RML	right lateral malleolus	anatomical	Right Lateral Malleolus
RMM	right medial malleolus	anatomical	Right Medial Malleolus
RMT1	right metatarso phalangeal joint 1	anatomical	Most Distal Aspect of the 1st Right Metatarsal Bone
RMT5	right metatarso phalangeal joint 5	anatomical	Most Distal Aspect of the 5th Right Metatarsal Bone
RPD1	right distal phalanx 1	anatomical	Centre of the 1st Right Toenail
RPSIS	right posterior superior iliac spine	anatomical	Prominence of right Posterior Superior Iliac Spine

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RS1	right shank 1	tracking	Bottom Edge of the Cluster is placed at the lateral Aspect of the Right Shank one hand Proximal of the Right Lateral Malleouls Marker
RS2	right shank 2	tracking	
RS3	right shank 3	tracking	
RS4	right shank 4	tracking	
RT1	right thigh 1	tracking	Bottom Edge of the Cluster is placed at the lateral Aspect of the Right Thigh one hand Proximal of the Right Lateral Epicondyle Marker
RT2	right thigh 2	tracking	
RT3	right thigh 3	tracking	
RT4	right thigh 4	tracking	
RTM	right trochanter major	anatomical	Prominence of the Right Major Trochanter
RTT	right tibial tuberosity	anatomical	Prominence of the Right Tibial Tuberosity
RWR	Right wrist radial	anatomical	Radial and most Distal Aspect of the Right Radius
RWU	Right wrist ulnar	anatomical	Dorsal and most Distal Aspect of the Right Ulna
UST	upper sternum	anatomical	Jugular Incisure
VC7	vertebra 7	anatomical	Prominence of the 7th Cervical Vertebra
V_JC_LAnkle	Virtual: left ankle joint centre	virtual	Arithmetic Mean of the LML and LMM
V_JC_LAnkle_Ground	Virtual: ground projection of V_JC_LAnkle	virtual	Ground Level Projection of the V_JC_LAnkle Marker
V_JC_LHip	Virtual: left hip joint centre	virtual	Marker calculated by means of Regression Formula of Hara et al., (2016)
V_JC_LKnee	Virtual: left knee joint centre	virtual	Arithmetic Mean of the LCL and LCM
V_JC_RAnkle	Virtual: right ankle joint centre	virtual	Arithmetic Mean of the RML and RMM

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V_JC_RAnkle_Ground	Virtual: ground projection of V_JC_RAnkle	virtual	Ground Level Projection of the V_JC_RAnkle Marker
V_JC_RHip	Virtual: right hip joint centre	virtual	Marker calculated by means of Regression Formula of Hara et al., (2016)
V_JC_RKnee	Virtual: right knee joint centre	virtual	Arithmetic Mean of the RCL and RCM
V_LCC_Ground	Virtual: ground projection of LCC	virtual	Ground Level Projection of the LCC Marker
V_LMid_Foot_Ground	Virtual: ground projection of arithmetic mean of LMT1 and LMT5	virtual	Ground Level Projection of the Arithmetic Mean of the LMT1 and LMT5 Markers
V_LMT1_Ground	Virtual: ground projection of LMT1	virtual	Ground Level Projection of the LMT1 Marker
V_LMT5_Ground	Virtual: ground projection of LMT5	virtual	Ground Level Projection of the LMT5 Marker
V_MidASIS	Virtual: mid ASIS	virtual	Arithmetic Mean of the RASIS and LASIS Markers
V_MidHJC	Virtual: mid hip joint	virtual	Arithmetic Mean of the V_JC_LHip and V_JC_RHip Markers
V_MidPelvis	Virtual: mid pelvis	virtual	Arithmetic Mean of the V_MidASIS and V_MidPSIS Markers
V_MidPSIS	Virtual: mid PSIS	virtual	Arithmetic Mean of the RPSIS and LPSIS Markers
V_RCC_Ground	Virtual: ground projection of RCC	virtual	Ground Level Projection of the RCC Marker
V_RMid_Foot_Ground	Virtual: ground projection of arithmetic mean of RMT1 and RMT5	virtual	Ground Level Projection of the Arithmetic Mean of the RMT1 and RMT5 Markers
V_RMT1_Ground	Virtual: ground projection of RMT1	virtual	Ground Level Projection of the RMT1 Marker

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V_RMT5_Ground	Virtual: ground projection of RMT5	virtual	Ground Level Projection of the RMT5 Marker

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Table 21: Average weighting of markers and coordinates across participants during the process of scaling

Marker	Weight
UST	10
VC7	10
LACJ	5
RACJ	5
LEL	1
LEM	1
LWU	1
LWR	1
REL	1
REM	1
RWU	1
RWR	1
LASIS	20
RASIS	20
LPSIS	20
RPSIS	20
LCL	2
LCM	2
LTT	15
LML	10
LMM	10
LCC	5
LMT1	15
LMT5	10
LPD1	5

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RCL	2
RCM	2
RTT	15
RML	10
RMM	10
RCC	5
RMT1	15
RMT5	10
RPD1	5

Coordinate	Weight	Value
Knee_angle_r	0.01	0
Mtp_angle_r	1	0
Knee_angle_l	0.01	0
Mtp_angle_l	1	0
Lumbar_extension	1	0
Arm_rot_r	1	0
Elbow_flex_r	1	0
Pro_sup_r	1	0
Arm_rot_l	1	0
Elbow_flex_l	1	0
Pro_sup_l	1	0

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Table 22: Overview of scale factors used during the OpenSim model scaling procedure

Name of Scale-Factor	Marker#1	Marker#2	Body	Axis
Torso_x	V_MidASIS	V_MidPSIS	torso	x
Torso_y	LACJ	LASIS	torso	y
	RACJ	RASIS		
	UST	V_MidASIS		
	VC7	V_MidPSIS		
Torso_z	LACJ	RACJ	torso	z
Pelvis_x	V_MidASIS	V_MidPSIS	pelvis	x
Pelvis_y	V_MidHJC	V_MidPelvis	pelvis	y
Pelvis_z	LASIS	RASIS	pelvis	z
	LPSIS	RPSIS		
Thigh_r_long	V_JC_RHip	V_JC_RKnee	Femur_r patella_r	y
Thigh_trans	RCL	RCM	Femur_r patella_r	x,z
Thigh_l_long	V_JC_LHip	V_JC_LKnee	Femur_l patella_l	y
Thigh_l_trans	LCL	LCM	Femur_l patella_l	x,z
Shank_r_long	V_JC_RKnee	V_JC_RAnkle	Tibia_r	y
Shank_r_trans	RML	RMM	Tibia_r	x,z
Shank_l_long	V_JC_LKnee	V_JC_LAnkle	Tibia_l	y
Shank_l_trans	LML	LMM	Tibia_l	x, z
Foot_r_long	V_RCC_Ground	V_RMid_Foot_Ground	Talus_r calcn_r	x
Foot_r_trans	V_RMT1_Ground	V_RMT5_Ground	Talus_r calcn_r toes_r	y, z

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Foot_l_long	V_LCC_Ground	V_LMid_Foot_Ground	Talus_l calcn_l	x
Foot_l_trans	V_LMT1_Ground	V_LMT5_Ground	Talus_l calcn_l toes_l	y, z
Toes_r_long	RPD1	RMT1	Toes_r	x
Toes_l_long	LMT1	LPD1	Toes_l	x
Upper_arm_r_long	RACJ	REM	Humerus_r	y
	RACJ	REL		
Lower_arm_r_long	REM	RWU	Ulna_r radius_r hand_r	y
	REL	RWR		
Upper_arm_l_long	LACJ	LEM	Humerus_l	y
	LACJ	LEL		
Lower_arm_l_long	LEM	LWU	Ulna_l radius_l hand_l	y
	LEL	LWR		
Arm_r_trans	REM	REL	Humerus_r ulna_r radius_r hand_r	x, z
Arm_l_trans	LEM	LEL	Humerus_l ulna_l radius_l hand_l	x, z

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Table 23: Average marker weights applied during IK for the right body side

Marker Name	Weight
RASIS	7.3
RPSIS	7.0
LASIS	43.3
LPSIS	43.3
LT2	3.o
RTT	17.3
LTT	1.6
TS2	5.0
LS2	1.7
RAJC	5.0
RCC	3.0
RMT1	2.0
RMT5	2.0
RPD1	0.02
LAJC	2.0
LCC	1.0
LMT1	1.5
LMT5	1.3
LPD1	1.5
UST	2.0
VC7	2.0
LACJ	5.0
RACJ	5.0
LHM	1.0
LEL	5.0
LEM	5.0

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LLAM	1.0
LWU	5.0
LWR	5.0
RHM	1.0
REL	5.0
REM	5.0
RLAM	1.0
RWR	5.0
RWU	5.0

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Table 24: Resistance bands and corresponding resistance ranges (Zen One Sports, 2026)

Resistance	Resistance Range
Yellow	2.3 – 4.5 kg
Green	4.5 – 6.8 kg
Red	6.8 – 9.1 kg
Blue	11.3 – 13.6 kg
Black	13.6 – 18.1 kg

Table 25: EHIS-PAQ results of study participants

Participant#	WRPA	TRPA	HEPA	MSPA	TPA
1	2	198	2	2	2
2	2	1956	1	2	1
3	2	1039,5	1	1	1
4	1	445,5	2	2	1
5	2	1548	1	1	1
6	2	462	1	1	1
7	2	3465	2	2	2
8	2	742,5	2	2	2
9	2	-	1	1	1
10	1	198	1	1	1
Reporting	**20%**	**1711,2**	**60%**	**50%**	**70%**

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Table 26: Repetition counts and resistance levels of the fatigue protocol, including summary statistics (mean, SD, median)

	Set 1		Set 2		Set 3		Set 4	
Participant#	**Res[a]**	**Rep[b]**	**Res**	**Rep**	**Res**	**Rep**	**Res**	**Rep**
1	Yellow	75	Green	18	Green	18	Green	10
2	Green	13	Green	10	Green	10	Green	9
3	Red	11	Red	12	Red	10	Red	9
4	Blue	15	Blue	11	Blue	10	Blue	8
5	Green	14	Green	13	Green	12	Green	13
6	Green	14	Green	9	Green	8	Green	10
7	Green	15	Green	13	Green	12	Green	11
8	Red	11	Red	8	Red	7	Green	10
9	Green	9	Green	9	Green	9	Green	7
10	Green	16	Green	13	Green	13	Green	12
Median	Green	14	Green	11.5	Green	10.0	Green	10.0
Mean		19.3		11.6		10.9		9.9
SD		19.7		2.9		3.1		1.8

[a] Colour-code of resistance band used

[b] Number of repetitions

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Table 27: Time intervals between completion of the fatigue protocol and post-fatigue motion capture and MVIC measurements

Participant#	end of fatigue protocol to post-fatigue-caption [a]	end of fatigue protocol to post-fatigue-MVIC of GMED [b]
1	not measured	not measured
2	65	220
3	45	900
4	45	295
5	68	240
6	160	170
7	80	320
8	20	300
9	38	not measured
10	31	325
Median	45	297,5
Mean	61,3	346
SD	41,6	230,2

[a], [b] transition time in [s]

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Appendix A

Intro Next I am going to ask you about the time you spend doing different types of physical activity in a typical week. Please answer these questions even if you do not consider yourself to be a physically active person.

Q1 Firstly think about the TIME you spend DOING WORK. Think of work as the things that you have to do such as paid and unpaid work, work around your home, taking care of family, studying or training.

When you are WORKING, which of the following best describes what you do?
Would you say …
Please refer your answer to the 'main work' you do. If you do multiple tasks, include all tasks. Select only one answer.

1 ◯ Mostly sitting or standing
2 ◯ Mostly walking or tasks of moderate physical effort
3 ◯ Mostly heavy labour or physically demanding work
4 ◯ Not performing any working tasks

Q2 The next questions EXCLUDE the WORK-RELATED PHYSICAL ACTIVITIES that you have already mentioned. Now I would like to ask you about the way you usually GET TO AND FROM PLACES. For example to work, to school, for shopping, or to market.

In a typical week, on how many days do you WALK for at least 10 minutes continuously in order to get to and from places?

1-7 Number of days: [ ]
0 ◯ Less than one day per week or never → *continue with Q4*

Q3 **How much time do you spend walking in order to get to and from places on a typical day?**

20 ◯ 10 - 29 minutes per day
45 ◯ 30 - 59 minutes per day
90 ◯ 1 hour to less than 2 hours per day
150 ◯ 2 hours to less than 3 hours per day
180 ◯ 3 hours or more per day

Q4 **In a typical week, on how many days do you BICYCLE for at least 10 minutes continuously to get to and from places?**

1-7 Number of days: [ ]
0 ◯ Less than one day per week or never → *continue with Q6*

Q5 **How much time do you spend bicycling in order to get to and from places on a typical day?**

20 ◯ 10 - 29 minutes per day
45 ◯ 30 - 59 minutes per day
90 ◯ 1 hour to less than 2 hours per day
150 ◯ 2 hours to less than 3 hours per day
180 ◯ 3 hours or more per day

Q6 The next questions EXCLUDE the WORK and TRANSPORT ACTIVITIES that you have already mentioned. Now I would like to ask you about SPORTS, FITNESS and RECREATIONAL (LEISURE) PHYSICAL ACTIVITIES that cause AT LEAST a small increase in breathing or heart rate. For example brisk walking, ball games, jogging, cycling or swimming.

In a typical week, on how many days do you carry out sports, fitness or recreational (leisure) activities for at least 10 minutes continuously?

1-7 Number of days: [ ]
0 ◯ Less than one day per week or never → *continue with Q8*

Q7 **How much time in total do you spend on sports, fitness or recreational (leisure) physical activities in a typical week?**

min [ ] : [ ] per week
hours minutes

Q8 **In a typical week, on how many days do you carry out activities specifically designed to STRENGTHEN your muscles such as doing resistance training or strength exercises? Include all such activities even if you have mentioned them before.**

1-7 Number of days: [ ]
0 ◯ Less than one day per week or never

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Table 1 Analyzing guideline for the European Health Interview Survey-Physical Activity Questionnaire, EHIS-PAQ

Indicator	Definition	Calculation	Label	Reporting
Work-related PA	Proportion of individuals doing mostly tasks with at least moderate physical effort when working	Generate WRPA = 1 if Q1 == 2 OR Q1 == 3; Replace WRPA = 2 if Q1 == 1; Replace WRPA = 3 if Q1 == 4	Label WRPA 'Mostly physically active when working' 1 = Yes 2 = No 3 = Not working	Percentage of individuals being mostly physically active when working (WRPA = 1)
Transport-related PA	Quintiles of transport-related physical activity in MET[a]/minutes per week	Enter data for Q3/Q5 as shown in Fig. 3; Generate WalkMin = Q2 x Q3; replace WalkMin = 0 if Q2 == 0 Generate CycleMin = Q4 x Q5; CylcleMin = 0 if Q4 == 0 GenerateTRPA = WalkMin x 3.3 + CycleMin x 6	Label TRPA 'Transport-related physical activity in MET/minutes per week'	Upper quintile limits of the transport-related physical activity index
Health-enhancing PA				
Aerobic PA guideline compliance	Proportion of individuals performing at least 150 min aerobic physical activity per week [b]	Enter data for Q7 in minutes per week; Replace Q7 = 0 if Q6 == 0; Generate AEROB = Q7 + CycleMin; Generate HEPA = 1 if AEROB ≥ 150; Replace HEPA = 2 if AEROB < 150[b]	Label HEPA 'Aerobic PA at least 150 min per week'[b] 1 = Yes 2 = No	Percentage of individuals performing at least 150 min aerobic physical activity per week[b] (HEPA = 1)
Muscle- strengthening PA guideline compliance	Proportion of individuals performing at least 2 days per week muscle-strengthening physical activity[c]	Generate MSPA = 1 if Q8 ≥ 2; Replace MSPA = 2 if Q8 < 2[c]	Label MSPA 'Muscle-strengthening physical activity at least 2 days per week'[c] 1 = Yes 2 = No	Percentage of individuals performing at least 2 days per week[c] muscle-strengthening physical activity (MSPA = 1)
Total PA	Proportion of individuals being sufficiently physically active in total	Generate TPA = 1 if HEPA == 1 OR WRPA == 1; Replace TPA = 2 if HEPA == 2 AND WRPA ≠ 1	Label TPA 'Sufficiently physically active in total' 1 = Yes 2 = No	Percentage of individuals being sufficiently physically active in total (TPA = 1)

[a]MET, metabolic equivalent (one MET corresponds to the energy expenditure at a state of complete rest). The MET values 3.3 for walking and 6 for cycling are adapted from the IPAQ-Long Form data analyzing guidelines [34]
[b]Cut-of-point is for the age group 18+ years, for the age group 15–17 years ≥ 60 min per day should be used
[c]Cut-of-point is for the age group 18+ years, for the age group 15–17 years ≥ 3 days per week should be used

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